

# Operating System Lab Report

## Introduction

The Linux operating system is a powerful and widely used open-source platform known for its flexibility, security, and command-line interface. Understanding basic and advanced Linux commands is essential for system administration, software development, and efficient file management.

This lab experiment is designed to provide hands-on experience with various Linux commands using the Ubuntu environment. The tasks include working with file and directory operations, user and permission management, system monitoring, compression techniques, and command-line utilities.

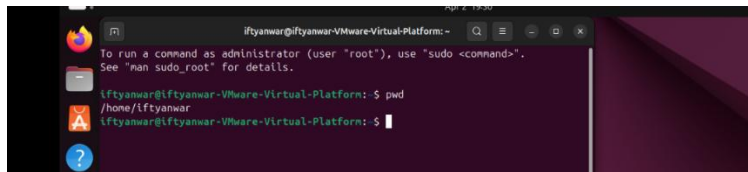
Through this lab, different commands were executed and observed to understand their functionality and real-life applications. This practical approach helps in developing problem-solving skills and improving familiarity with the Linux terminal.

### 1. Command: pwd

Syntax:

Pwd

Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' showing the execution of the 'pwd' command. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~\$'. The command 'pwd' is entered, and the output is '/home/iftyanwar'. The prompt then changes to 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~\$'.

Explanation:

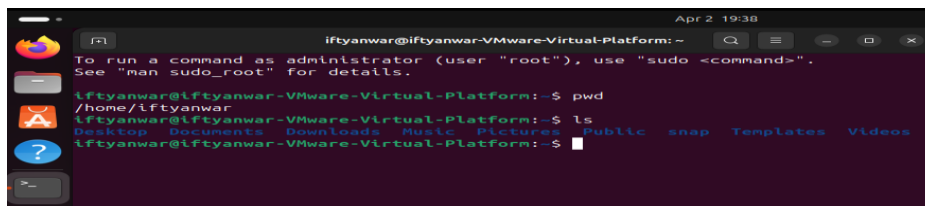
The pwd command helps the user to identify the current location in the file system by showing the full path of the working directory.

### 2 Command: ls

Syntax:

ls

Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' showing the execution of the 'ls' command. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~\$'. The command 'ls' is entered, and the output is 'Desktop Documents Downloads Music Pictures Public snap Templates Videos'. The prompt then changes to 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~\$'.

Explanation:

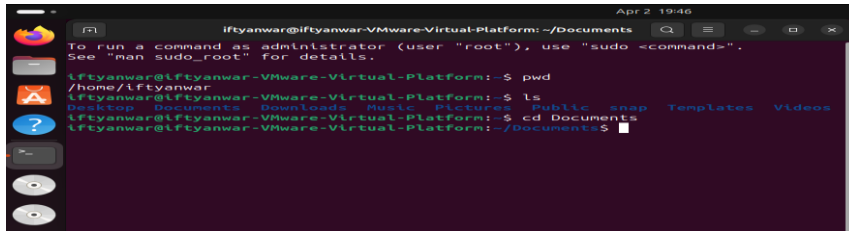
The ls command is used to display a list of all files and directories in the current working directory. It helps the user to view and identify available folders and files in that location.

### 3 Command: cd

#### Syntax:

cd Documents

#### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/Documents' showing the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ pwd
/home/iftyanwar
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cd Documents
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/Documents$
```

#### Explanation:

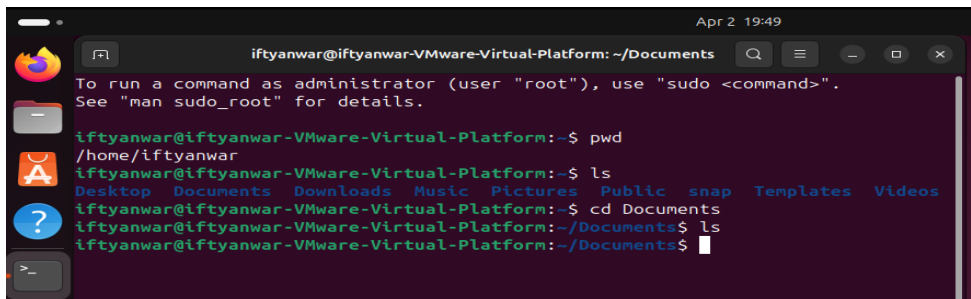
The cd command is used for navigating between directories in the Linux file system. By using cd Documents, the user changes the current location to the Documents folder, making it possible to work with its contents.

### 4 Command: ls

#### Syntax:

ls

#### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/Documents' showing the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ pwd
/home/iftyanwar
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cd Documents
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/Documents$ ls
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/Documents$
```

#### Explanation:

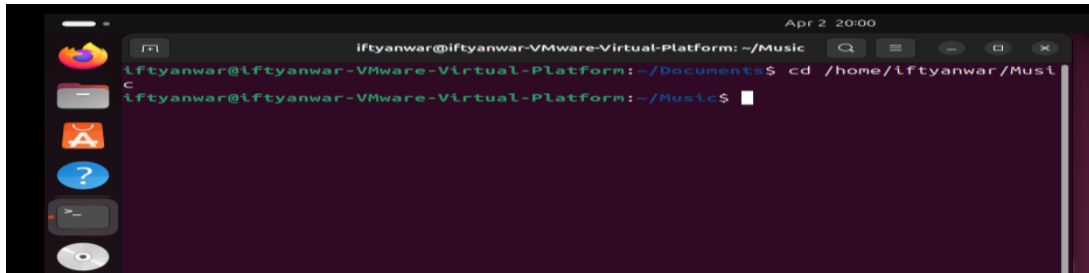
The ls command lists all files and directories in the current working directory. Here, it shows no output, indicating that the Documents folder does not contain any files or subdirectories.

### 5 Command: cd /home/user/Music

#### Syntax:

cd /home/iftyanwar/Music

#### Output:



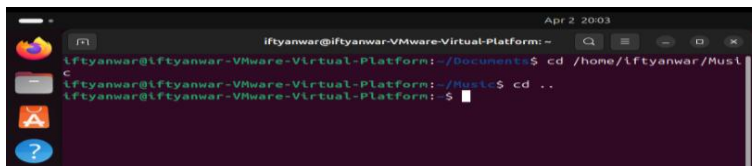
The `cd` command helps the user navigate through directories. By using the full path `/home/iftyanwar/Music`, it directly takes the user to the Music folder, making it easier to access its contents from any location.

**6 Command: `cd ..`**

**Syntax:**

`cd ..`

**Output:**



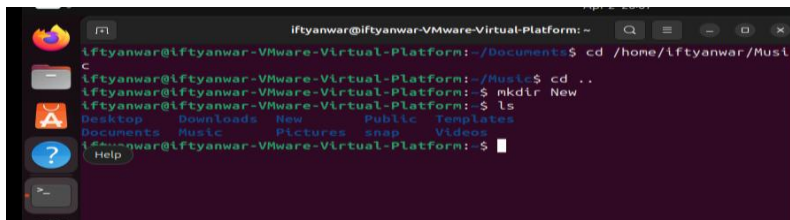
**Explanation:**

The `cd ..` command allows the user to go back to the parent directory from the current location. Here, it moves the user from the Music folder to the home directory, helping in easy navigation within the file system.

**Syntax:**

`mkdir New`

**Output:**



**Explanation:**

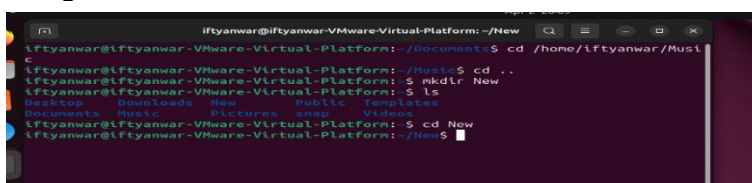
The `mkdir` command allows the user to create a new folder in the current directory. Here, it successfully creates a directory named "New", which can be used for organizing files or storing data.

**8 Command: `cd New`**

**Syntax:**

`cd New`

**Output:**



### Explanation:

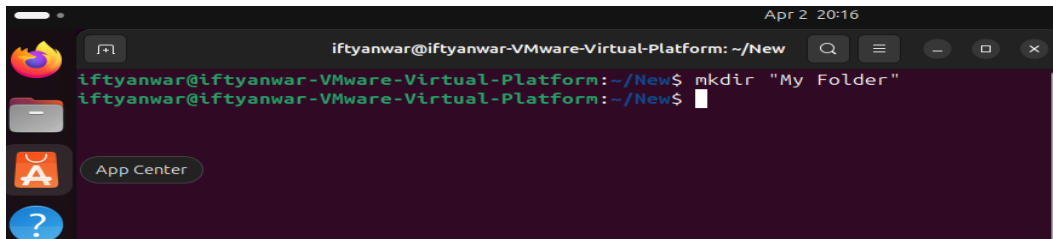
The `cd` command helps the user navigate into different directories. Here, it takes the user into the "New" folder, allowing access to its contents and enabling further operations inside it.

## 9 Command: mkdir (Two Words)

### Syntax:

`mkdir "My Folder"`

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ mkdir "My Folder"
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$
```

### Explanation:

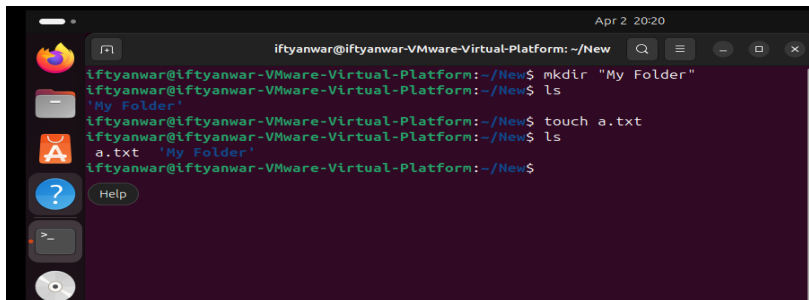
The `mkdir` command helps create a new folder in the current directory. In this case, quotation marks are used to include a space in the folder name, allowing the creation of a directory named "My Folder".

## 10 Command: touch

### Syntax:

`touch a.txt`

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ mkdir "My Folder"
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ ls
'My Folder'
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ touch a.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ ls
a.txt 'My Folder'
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$
```

### Explanation:

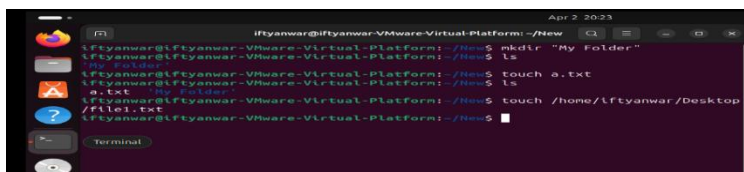
The `touch` command allows the user to create a new empty file in the current working directory. Here, it successfully creates a file named "a.txt", which can later be used to store data.

## 11 Command: touch (with path)

### Syntax:

`touch /home/iftyanwar/Desktop/file1.txt`

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ mkdir "My Folder"
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ ls
'My Folder'
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ touch a.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ ls
a.txt 'My Folder'
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ touch /home/iftyanwar/Desktop/file1.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$
```

**Explanation:**

The touch command allows the user to create a file in any desired location by specifying the full path. Here, it creates a file named "file1.txt" directly on the Desktop, making it easy to access.

**12 Command: nano****Syntax:**

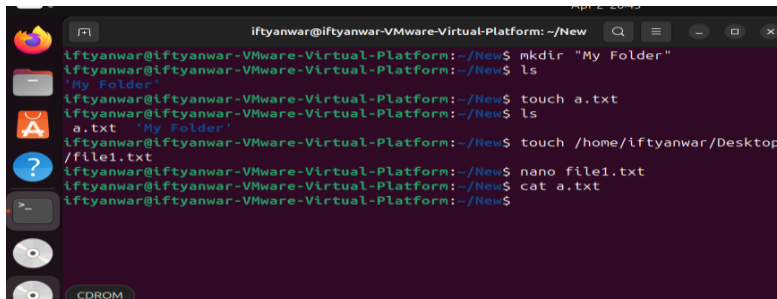
nano file1.txt

**Output:****Explanation:**

The nano command provides a simple text editor inside the terminal. It allows the user to write or edit text in a file easily, making it useful for quick file editing tasks.

**13 Command: cat****Syntax:**

cat a.txt

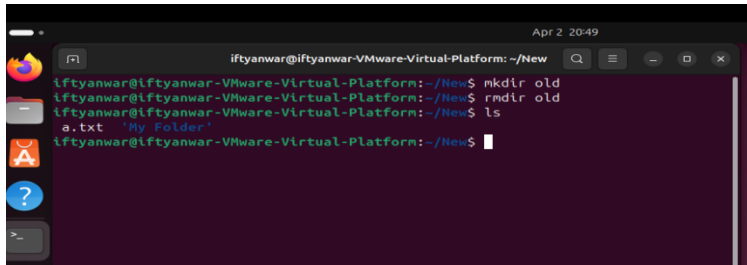
**Output:****Explanation:**

The cat command helps the user view the contents of a file directly in the terminal. Here, no content is displayed because the file "a.txt" is empty.

**14 Command: rmdir****Syntax:**

rmdir old

**Output:**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New' with a timestamp of 'Apr 2 20:49'. The terminal shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ mkdir old
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ rmdir old
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ ls
a.txt  'My Folder'
```

**Explanation:**

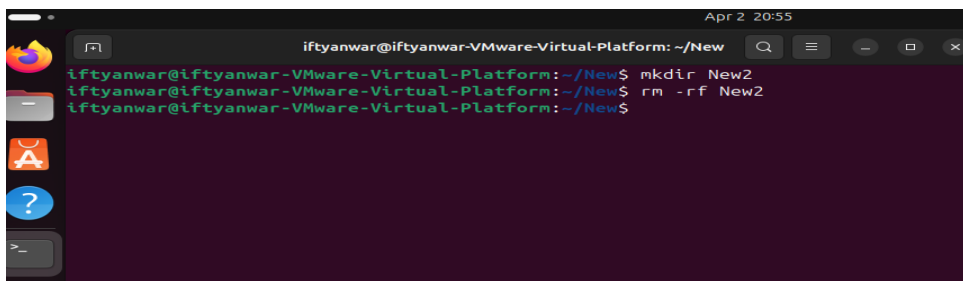
The rmdir command allows the user to delete an empty folder from the system. Here, it removes the directory named "old", which helps in keeping the file system organized.

**15 Command: rm -rf**

**Syntax:**

rm -rf New2

**Output:**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New' with a timestamp of 'Apr 2 20:55'. The terminal shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ mkdir New2
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ rm -rf New2
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$
```

**Explanation:**

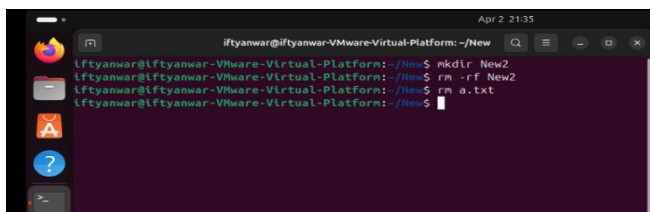
The rm -rf command allows the user to permanently delete a folder along with all its contents. It works forcefully without any confirmation, making it a powerful but potentially dangerous command.

**16 Command: rm**

**Syntax:**

rm a.txt

**Output:**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New' with a timestamp of 'Apr 2 21:35'. The terminal shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ mkdir New2
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ rm -rf New2
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ rm a.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$
```

**Explanation:**

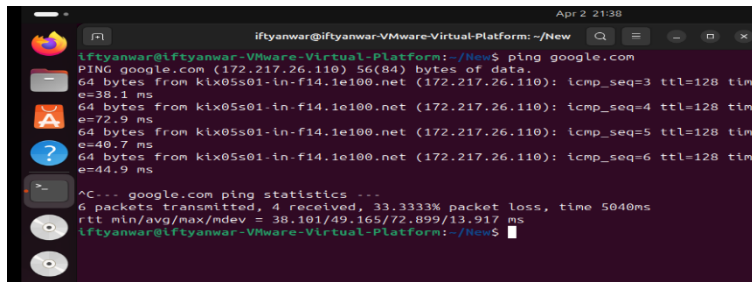
The rm command allows the user to delete files from the current directory. Here, it removes the file "a.txt", helping to manage and clean unnecessary files.

**17 Command: ping**

**Syntax:**

ping google.com

**Output:**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New' showing the execution of the 'ping google.com' command. The output displays four successful ping attempts with varying response times (e.g., 38.1 ms, 72.9 ms, 40.7 ms, 44.9 ms). Below the individual pings, a summary line shows '6 packets transmitted, 4 received, 33.333% packet loss, time 5040ms' and 'rtt min/avg/max/mdev = 38.101/49.165/72.899/13.917 ms'.

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ ping google.com
PING google.com (172.217.26.110) 56(84) bytes of data:
64 bytes from kix05s01-in-f14.1e100.net (172.217.26.110): icmp_seq=3 ttl=128 tim
e=38.1 ms
64 bytes from kix05s01-in-f14.1e100.net (172.217.26.110): icmp_seq=4 ttl=128 tim
e=72.9 ms
64 bytes from kix05s01-in-f14.1e100.net (172.217.26.110): icmp_seq=5 ttl=128 tim
e=40.7 ms
64 bytes from kix05s01-in-f14.1e100.net (172.217.26.110): icmp_seq=6 ttl=128 tim
e=44.9 ms
^C... google.com ping statistics ...
6 packets transmitted, 4 received, 33.333% packet loss, time 5040ms
rtt min/avg/max/mdev = 38.101/49.165/72.899/13.917 ms
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$
```

### Explanation:

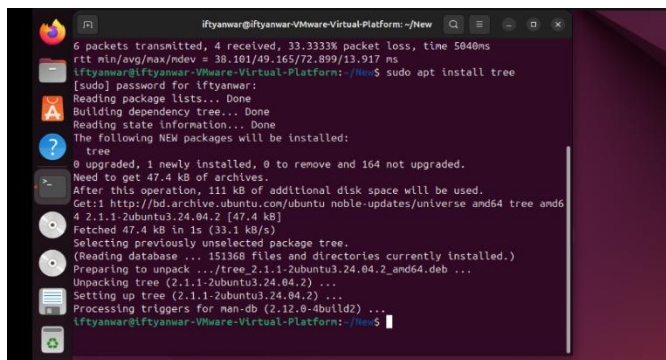
The ping command helps the user test internet connectivity by sending data packets to a server (like google.com). It shows whether the connection is active and how fast the response time is.

### 18 Command: sudo apt install

#### Syntax:

sudo apt install tree

#### Output:

A terminal window showing the execution of 'sudo apt install tree'. The output indicates that the package 'tree' is being installed. It shows the package list being read, the dependency tree being built, and the state information being read. The following NEW packages will be installed: tree. It also shows the disk space requirements and the fetching of the package from the repository. The installation is successful.

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ sudo apt install tree
[sudo] password for iftyanwar:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
tree
0 upgraded, 1 newly installed, 0 to remove and 164 not upgraded.
Need to get 47.4 kB of archives.
After this operation, 111 kB of additional disk space will be used.
Get:1 http://bd.archive.ubuntu.com/ubuntu noble-updates/universe amd64 tree amd6
4 2.1.1-2ubuntu3.24.04.2 [47.4 kB]
Fetched 47.4 kB in 1s (33.1 kB/s)
Selecting previously unselected package tree.
(Reading database ... 151368 files and directories currently installed.)
Preparing to unpack .../tree_2.1.1-2ubuntu3.24.04.2_amd64.deb ...
Unpacking tree (2.1.1-2ubuntu3.24.04.2) ...
Setting up tree (2.1.1-2ubuntu3.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$
```

### Explanation:

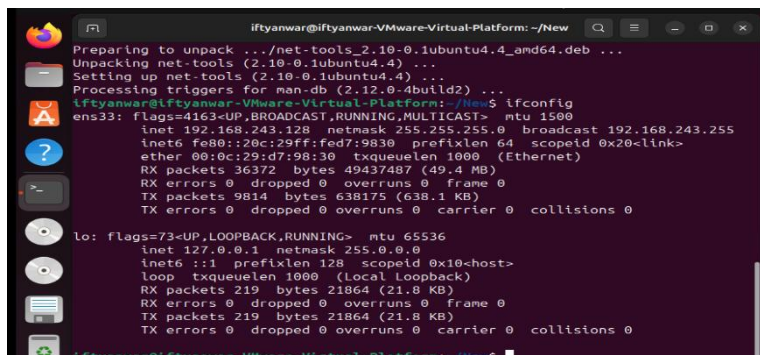
The sudo apt install command allows the user to install new software packages in the system. By using sudo, administrative permission is granted, and the "tree" package is successfully installed for viewing directory structures.

### 19 Command: ifconfig

#### Syntax:

ifconfig

#### Output:

A terminal window showing the execution of the 'ifconfig' command. The output displays network configuration details for the 'ens33' and 'lo' interfaces. For 'ens33', it shows flags, IP address (192.168.243.128), netmask (255.255.255.0), broadcast address (192.168.243.255), MAC address (08:00:27:0d:79:83), and statistics for RX and TX packets. For 'lo', it shows flags, IP address (127.0.0.1), netmask (255.0.0.0), and statistics for RX and TX packets.

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.243.128 netmask 255.255.255.0 broadcast 192.168.243.255
    inet6 fe80::20c:29ff:fed7:9830 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:0d:79:83 txqueuelen 1000 (Ethernet)
    RX packets 36372 bytes 49437487 (49.4 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9814 bytes 638175 (638.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 219 bytes 21864 (21.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 219 bytes 21864 (21.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$
```

### Explanation

The `ifconfig` command is used to display detailed information about the network interfaces of a system. It shows important data such as the IP address, subnet mask, broadcast address, and MAC address of each network interface. This command also provides information about the status of the network (whether it is running or not) and data transmission statistics.

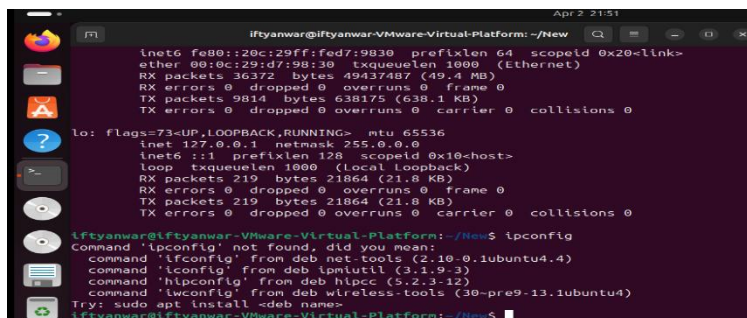
In modern Ubuntu systems, the `ifconfig` command is not installed by default. Therefore, the `net-tools` package needs to be installed before using this command. After installation, it helps users easily monitor and troubleshoot network-related information

## 20 Command: `ipconfig`

### Syntax:

`ipconfig`

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New
ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 219 bytes 21864 (21.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ ipconfig
Command 'ipconfig' not found, did you mean:
  command 'ifconfig' from deb net-tools (2.10-0.1ubuntu4.4)
  command 'lconfig' from deb ipmitool (3.1.9-3)
  command 'hupconfig' from deb hupcc (5-2.3-12)
  command 'lwconfig' from deb wireless-tools (30-pre9-13.1ubuntu4)
Try: sudo apt install <deb name>
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$
```

### Explanation

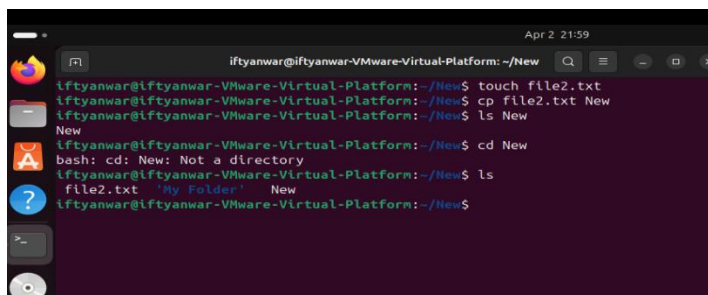
The `ipconfig` command is used in Windows operating systems to display network configuration details such as IP address, subnet mask, and default gateway. However, this command is not available in Linux systems like Ubuntu. When executed in Ubuntu, it shows a “command not found” error. In Linux, similar functionality is provided by commands such as `ifconfig` or `ip`.

## 21 Command: `cp`

### Syntax:

`cp file2.txt New`

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ touch file2.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ cp file2.txt New
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ ls New
New
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ cd New
bash: cd: New: Not a directory
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$ ls
file2.txt 'My Folder' New
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/New$
```

### Explanation

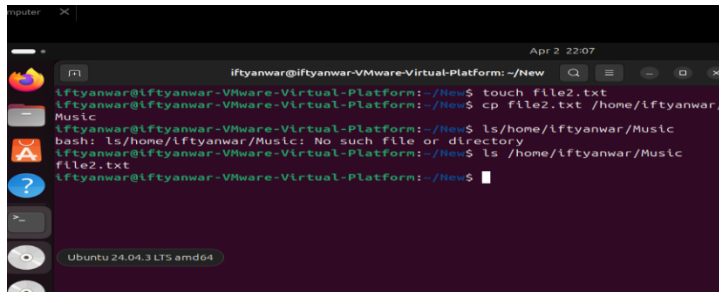
The `cp` command is used to copy files from one location to another. In this case, an error occurred because “New” was not a directory but a file. As a result, the system could not copy the file into it. This highlights the importance of ensuring that the destination is a valid directory before performing the copy operation.

## 22 Command: cp (with path)

### Syntax:

cp file2.txt /home/iftyanwar/Music

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ touch file2.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ cp file2.txt /home/iftyanwar/Music
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ ls /home/iftyanwar/Music
bash: ls/home/iftyanwar/Music: No such file or directory
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$ ls /home/iftyanwar/Music
file2.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform:~/New$
```

### Explanation

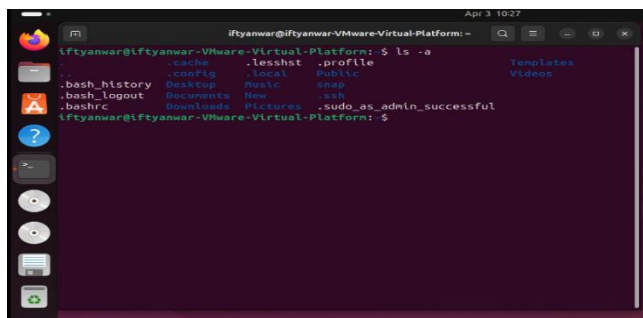
The cp command is used to copy files from one location to another. In this example, the file file2.txt is copied from the current directory to the Music directory using an absolute path. This ensures that the file is placed in the exact destination regardless of the current working directory.

## 23 Command: ls -a

### Syntax:

ls -a

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' shows the command 'ls -a' and its output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls -a
.  .cache  .lessht  .profile  Templates  Videos
.bash_history  .config  .local  Public
.bash_logout  Desktop  Music  snap
.bashrc  Documents  New  ssh
Downloads  Pictures  .sudo_as_admin_successful
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

### Explanation:

The ls -a command is used to display all files and directories, including hidden files. In Linux, hidden files start with a dot (.). This command shows both normal and hidden files in the current directory. The . represents the current directory, and .. represents the parent directory.

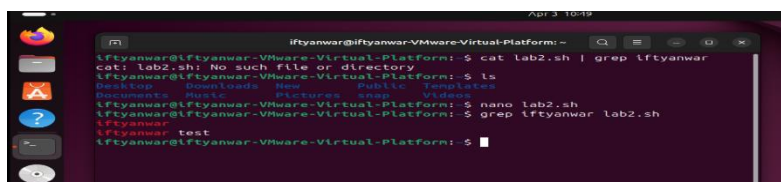
## 24 Command: grep

Use Case: Search text

### Syntax:

grep iftyanwar lab2.sh

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cat lab2.sh | grep iftyanwar
cat: lab2.sh: No such file or directory
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls
Desktop  Downloads  New  Public  Templates  Videos
Documents  Music  Pictures  snap
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ nano lab2.sh
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ grep iftyanwar lab2.sh
iftyanwar
test
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

## Explanation:

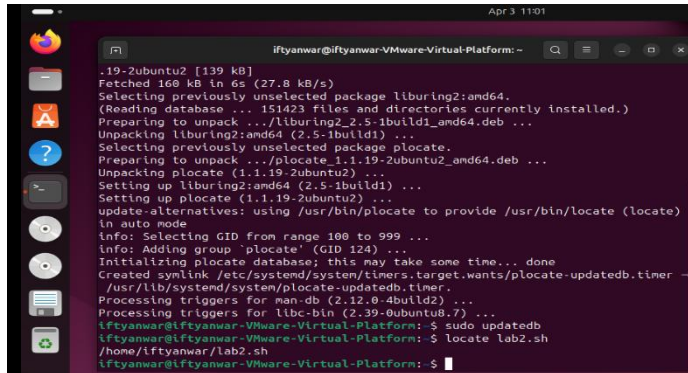
The grep command is used to search for specific text inside a file. In this example, it searches for the word "iftyanwar" in the file lab2.sh and displays all matching lines.

## 25 Command: locate

### Syntax:

locate lab2.sh

### Output:



```
Apr 3 11:01
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
.19-2ubuntu2 [139 kB]
Fetched 160 kB in 6s (27.8 kB/s)
Selecting previously unselected package liburing2:amd64.
(Reading database ... 151423 files and directories currently installed.)
Preparing to unpack .../liburing2_2.5-1build1_amd64.deb ...
Unpacking liburing2:amd64 (2.5-1build1) ...
Selecting previously unselected package plocate.
Preparing to unpack .../plocate_1.1.19-2ubuntu2_amd64.deb ...
Unpacking plocate (1.1.19-2ubuntu2) ...
Setting up liburing2:amd64 (2.5-1build1) ...
Setting up plocate (1.1.19-2ubuntu2) ...
update-alternatives: using /usr/bin/plocate to provide /usr/bin/locate (locate)
in auto mode
info: Selecting GID from range 100 to 999 ...
info: Adding group 'plocate' (GID 124) ...
Initializing plocate database; this may take some time... done
Created symlink /etc/systemd/system/timers.target.wants/plocate-updatedb.timer →
/usr/lib/systemd/system/plocate-updatedb.timer.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ sudo updatedb
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ locate lab2.sh
/home/iftyanwar/lab2.sh
iftyanwar@iftyanwar-VMware-Virtual-Platform: $
```

### Explanation

The locate command is used to quickly search for files using a pre-built database. It returns the full path of the file, making it easier to locate files in the system. It is faster than the find command because it does not search the entire filesystem in real time.

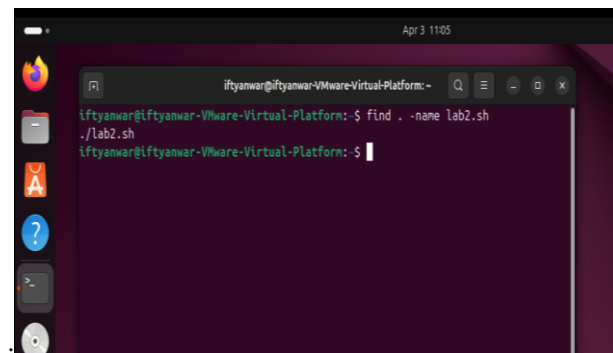
## 26 Command: find (name)

Use Case: Find exact file

### Syntax:

find . -name lab2.sh

### Output:



```
Apr 3 11:05
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ find . -name lab2.sh
./lab2.sh
iftyanwar@iftyanwar-VMware-Virtual-Platform: $
```

### Explanation

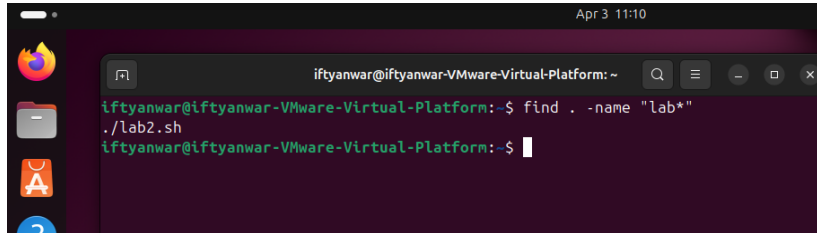
The find command is used to search for files and directories. In this example, it searches for a file named "lab2.sh" in the current directory (.) and returns its exact location.

## 27 Command: find (wildcard)

### Syntax:

```
find . -name "lab*"
```

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a search icon and window controls. It shows the command 'find . -name "lab\*"' being executed, which returns './lab2.sh' as the only result.

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ find . -name "lab*"  
./lab2.sh  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

### Explanation

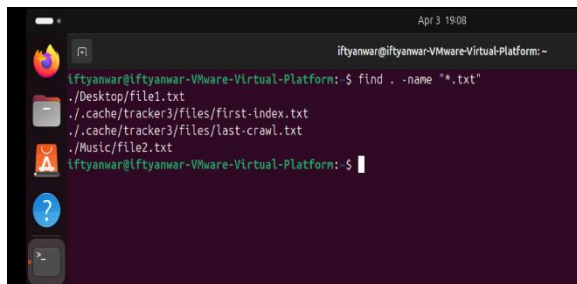
The find command with wildcard (\*) is used to search for files based on patterns. In this example, it finds all files that start with "lab" in the current directory

## 28 Command: find (.txt files)

### Syntax:

```
find . -name "*.txt"
```

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' showing the command 'find . -name "\*.txt"'. The output lists several text files with their full paths: './Desktop/file1.txt', './.cache/tracker3/files/first-index.txt', './.cache/tracker3/files/last-crawl.txt', and './Music/file2.txt'.

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ find . -name "*.txt"  
./Desktop/file1.txt  
./.cache/tracker3/files/first-index.txt  
./.cache/tracker3/files/last-crawl.txt  
./Music/file2.txt  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

### Explanation

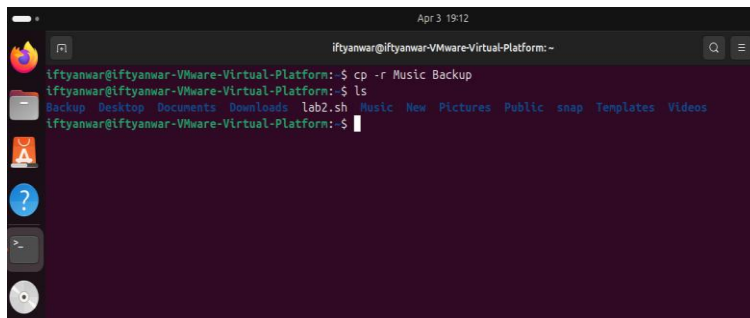
The find command with ".txt" is used to search for all text files in a directory. The wildcard (\*) matches any filename that ends with ".txt". It searches recursively from the current directory and shows the full path of each file.

## 29 Command: cp -r

### Syntax:

```
cp -r Music Backup
```

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' showing the command 'cp -r Music Backup' being executed. After the command, the user runs 'ls' to show the directory contents, which now includes 'Backup' along with other folders like 'Desktop', 'Documents', etc.

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cp -r Music Backup  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls  
Backup Desktop Documents Downloads lab2.sh Music New Pictures Public snap Templates Videos  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

## Explanation

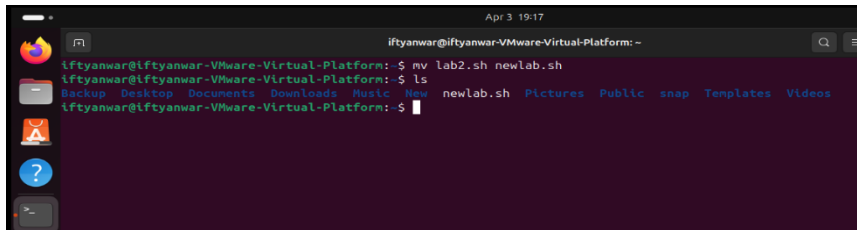
The cp -r command is used to copy directories along with all their contents. The -r (recursive) option ensures that all files and subdirectories inside the source directory are copied to the destination directory.

### 30 Command: mv (rename)

#### Syntax:

```
mv lab2.sh newlab.sh
```

#### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a date of 'Apr 3 19:17'. The user enters the command 'mv lab2.sh newlab.sh'. The prompt changes to 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'. The user then enters 'ls', and the output shows 'Backup Desktop Documents Downloads Music New newlab.sh Pictures Public snap Templates Videos'. The prompt returns to 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'.

## Explanation

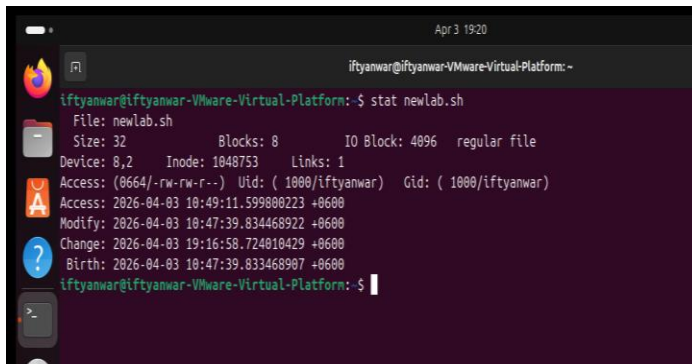
The mv command is used to move or rename files. In this example, the file "lab2.sh" is renamed to "newlab.sh". The original file name is replaced with the new name

### 31 Command: stat

#### Syntax:

```
stat newlab.sh
```

#### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a date of 'Apr 3 19:20'. The user enters the command 'stat newlab.sh'. The output displays file metadata: 'File: newlab.sh', 'Size: 32', 'Blocks: 8', 'IO Block: 4096', 'regular file', 'Device: 8,2', 'Inode: 1048753', 'Links: 1', 'Access: (0664/-rw-rw-r--) Uid: ( 1000/iftyanwar) Gid: ( 1000/iftyanwar)', 'Access: 2026-04-03 10:49:11.599800223 +0600', 'Modify: 2026-04-03 10:47:39.834468922 +0600', 'Change: 2026-04-03 19:16:58.724010429 +0600', and 'Birth: 2026-04-03 10:47:39.833468907 +0600'. The prompt returns to 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'.

## Explanation

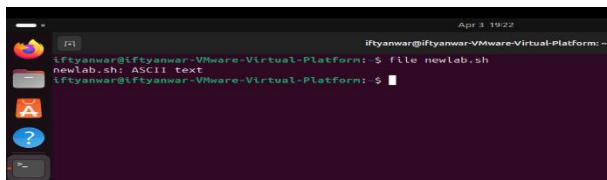
The stat command is used to display detailed information about a file. It shows file size, permissions, last modified time, and other metadata.

### 32 Command: file

#### Syntax:

```
file newlab.sh
```

#### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a date of 'Apr 3 19:22'. The user enters the command 'file newlab.sh'. The output shows 'newlab.sh: ASCII text'. The prompt returns to 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'.

### Explanation

The file command is used to determine the type of a file. It analyzes the file content and tells whether it is a text file, script, binary file, etc. In this example, it shows that the file is an ASCII text file.

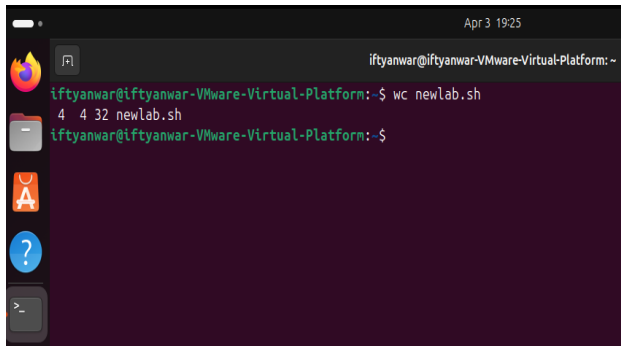
### 33 Command: wc

**Use Case:** Count lines, words, characters

#### Syntax:

wc newlab.sh

#### Output:

A terminal window with a dark purple background. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~'. The command 'wc newlab.sh' has been entered and executed. The output is '4 4 32 newlab.sh'. The terminal window has a title bar that says 'Apr 3 19:25' and a sidebar with icons for applications like Firefox, Files, and the Dash menu.

### Explanation

The wc command is used to count the number of lines, words, and characters in a file.

In the **output**:

First number = lines

Second number = words

Third number = characters

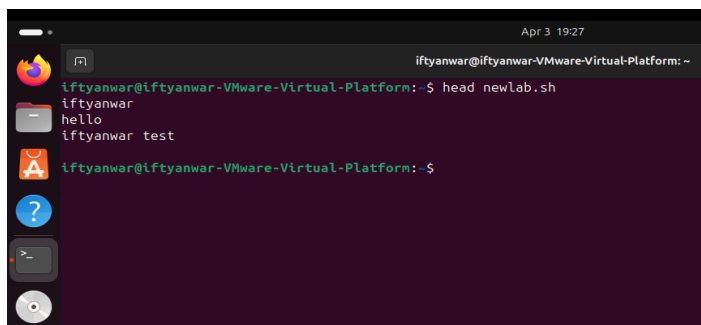
### 34 Command: head

**Use Case:** Show first lines of file

#### Syntax:

head newlab.sh

#### Output:

A terminal window with a dark purple background. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~'. The command 'head newlab.sh' has been entered and executed. The output shows the first three lines of the file: 'iftyanwar', 'hello', and 'iftyanwar test'. The terminal window has a title bar that says 'Apr 3 19:27' and a sidebar with icons for applications like Firefox, Files, and the Dash menu.

### Explanation

The head command is used to display the first few lines of a file. By default, it shows the first 10 lines, which helps users quickly view the beginning of a file.

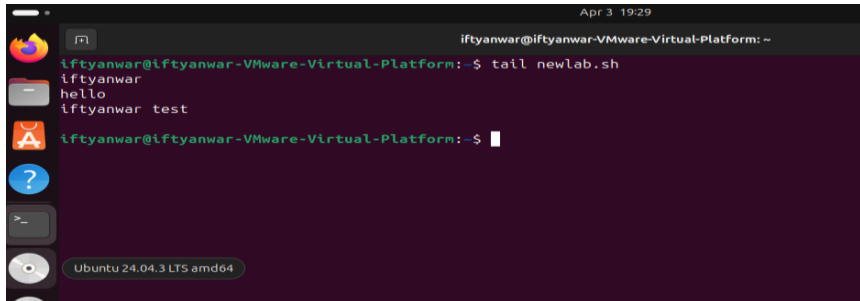
### 35 Command: tail

**Use Case:** Show last lines of file

#### **Syntax:**

tail newlab.sh

**Output :**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a date/time stamp of 'Apr 3 19:29'. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'. The user enters 'tail newlab.sh'. The output shows the last three lines of the file 'newlab.sh': 'iftyanwar', 'hello', and 'iftyanwar test'. The prompt returns to 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'.

#### **Explanation**

The tail command is used to display the last few lines of a file. By default, it shows the last 10 lines, which helps users quickly check the end of a file.

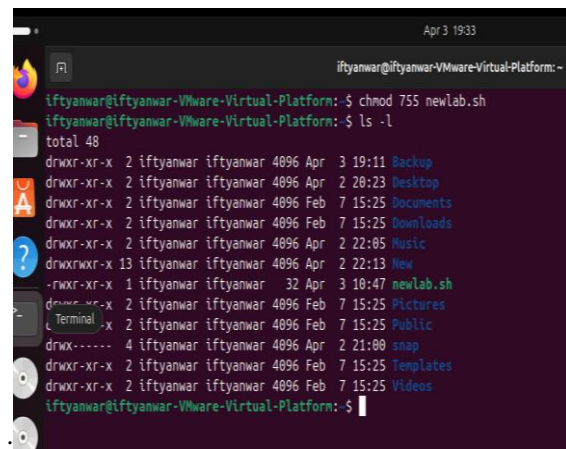
### 36 Command: chmod

**Use Case:** Change file permissions

#### **Syntax:**

chmod 755 newlab.sh

**Output :**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a date/time stamp of 'Apr 3 19:33'. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'. The user enters 'chmod 755 newlab.sh'. The prompt returns to 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'. The user then enters 'ls -l'. The output shows a directory listing with permissions, owner, group, size, date, and filename. The file 'newlab.sh' is listed with permissions 'drwxr-xr-x', owner 'iftyanwar', group 'iftyanwar', size '32', date 'Apr 3 10:47', and filename 'newlab.sh'. The prompt returns to 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'.

#### **Explanation**

The chmod command is used to change file permissions.

Here, 755 means:

**7 (Owner)** → read, write, execute

**5 (Group)** → read, execute

**5 (Others)** → read, execute

After applying this command, the file newlab.sh becomes executable.

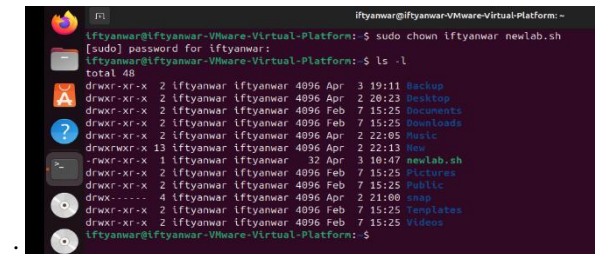
### 37 Command: chown

Use Case: Change file owner

#### Syntax:

sudo chown iftyanwar newlab.sh

#### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
$ sudo chown iftyanwar newlab.sh  
[sudo] password for iftyanwar:  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ ls -l  
total 48  
drwxr-xr-x 2 iftyanwar iftyanwar 4096 Apr 3 19:11 backup  
drwxr-xr-x 2 iftyanwar iftyanwar 4096 Apr 2 20:23 Desktop  
drwxr-xr-x 2 iftyanwar iftyanwar 4096 Feb 7 15:25 Documents  
drwxr-xr-x 2 iftyanwar iftyanwar 4096 Feb 7 15:25 Downloads  
drwxr-xr-x 13 iftyanwar iftyanwar 4096 Apr 2 22:05 Music  
drwxr-xr-x 1 iftyanwar iftyanwar 32 Apr 3 10:47 newlab.sh  
drwxr-xr-x 2 iftyanwar iftyanwar 4096 Feb 7 15:25 Pictures  
drwxr-xr-x 2 iftyanwar iftyanwar 4096 Feb 7 15:25 Public  
drwxr-xr-x 4 iftyanwar iftyanwar 4096 Apr 2 21:00 snap  
drwxr-xr-x 2 iftyanwar iftyanwar 4096 Feb 7 15:25 Templates  
drwxr-xr-x 2 iftyanwar iftyanwar 4096 Feb 7 15:25 Videos  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $
```

#### Explanation

The chown command is used to change the ownership of a file.

In this example, the file newlab.sh is assigned to the user **iftyanwar**.

The sudo command is required because changing file ownership needs administrative privileges.

After executing the command, the ownership is verified using ls -l.

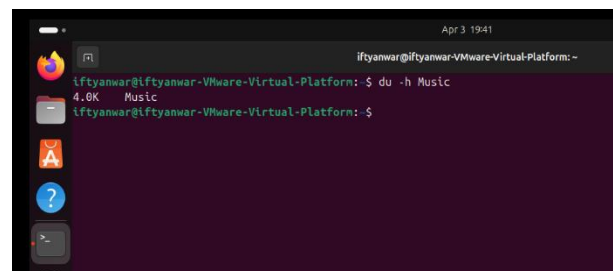
### 38 Command: du

Use Case: Check directory size

#### Syntax:

du -h Music

#### Output:



```
Apr 3 19:41  
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
$ du -h Music  
4.0K Music  
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
```

#### Explanation

The du (disk usage) command is used to check the size of files and directories.

The -h option displays the size in a human-readable format (KB, MB, GB).

Here, it shows how much space the Music directory is using.

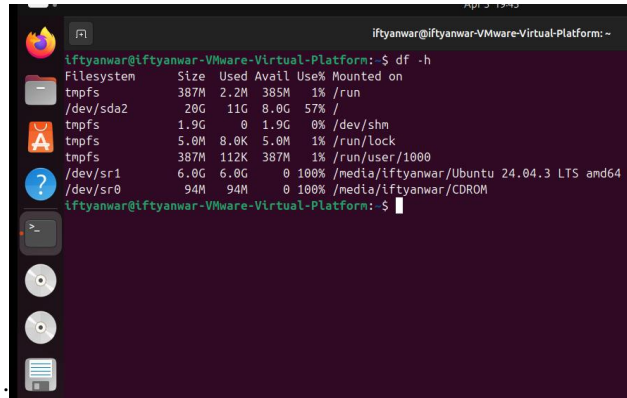
### 39 Command: df

**Use Case:** Check disk space

**Syntax:**

df -h

**Output:**



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs           387M  2.2M  385M   1% /run  
/dev/sda2       20G   11G   8.0G  57% /  
tmpfs           1.9G   0 1.9G   0% /dev/shm  
tmpfs           5.0M   8.0K  5.0M   1% /run/lock  
tmpfs           387M  112K  387M   1% /run/user/1000  
/dev/sr1        6.0G   6.0G   0 100% /media/iftyanwar/Ubuntu 24.04.3 LTS amd64  
/dev/sr0        94M   94M   0 100% /media/iftyanwar/CDROM  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation**

The df (disk free) command is used to check disk space usage of all mounted filesystems.

The -h option displays the output in a human-readable format (GB, MB).

It shows total size, used space, available space, and usage percentage.

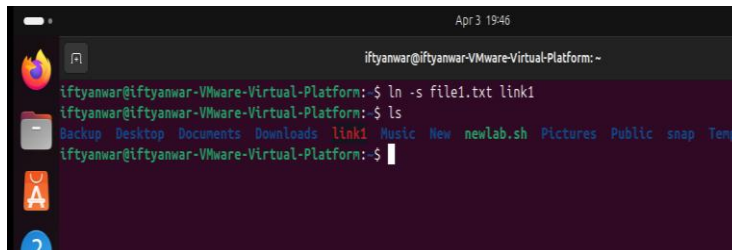
### 40 Command: ln -s

**Use Case:** Create symbolic link

**Syntax:**

ln -s file1.txt link1

**Output:**



```
Apr 3 19:46  
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ln -s file1.txt link1  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls  
Backup Desktop Documents Downloads link1 Music New newlab.sh Pictures Public snap Temp  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation**

The ln -s command is used to create a symbolic (soft) link.

A symbolic link acts like a shortcut to another file.

In this example, link1 is created as a link pointing to file1.txt.

The ls command confirms that the link has been created successfully

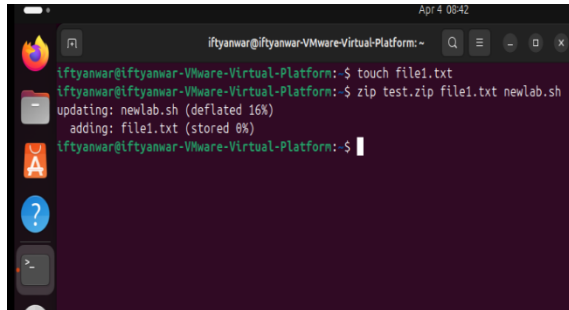
#### 41 Command: zip

**Use Case:** Compress files

**Syntax:**

zip test.zip file1.txt newlab.sh

**Output:**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a search bar and window controls. The terminal shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$ touch file1.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$ zip test.zip file1.txt newlab.sh
updating: newlab.sh (deflated 16%)
adding: file1.txt (stored 0%)
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$
```

**Explanation**

The zip command is used to compress multiple files into a single archive.

In this example, both file1.txt and newlab.sh are successfully added to test.zip.

The message deflated indicates compression, while stored means no compression was needed.

Since both files exist, the command executed successfully without any errors

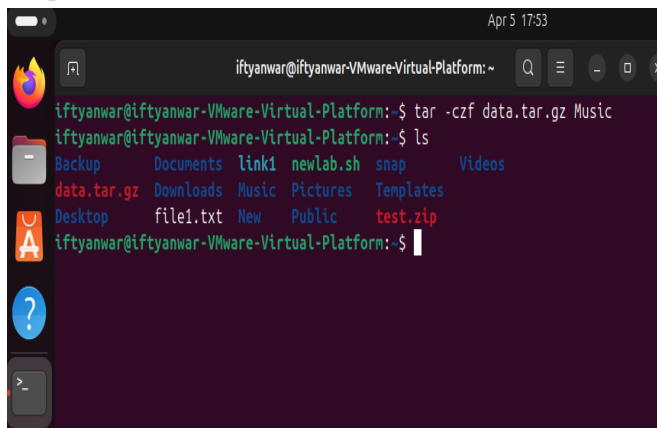
#### 42 Command: tar -czf

**Use Case:** Create compressed archive

**Syntax:**

tar -czf data.tar.gz Music

**Output :**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a search bar and window controls. The terminal shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$ tar -czf data.tar.gz Music
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$ ls
Backup  Documents  link1  newlab.sh  snap  Videos
data.tar.gz  Downloads  Music  Pictures  Templates
Desktop  file1.txt  New    Public    test.zip
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$
```

**Explanation**

The tar -czf command is used to create a compressed archive using gzip.

Here, the Music directory is compressed into a file named data.tar.gz.

The -c option creates the archive, -z enables gzip compression, and -f specifies the filename.

The ls command confirms that the archive file has been created successfully.

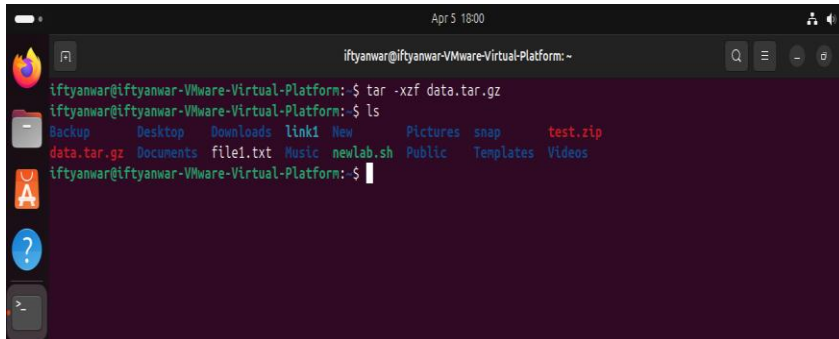
#### 43 Command: tar -xzf

**Use Case:** Extract compressed archive

**Syntax:**

`tar -xzf data.tar.gz`

**Output :**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a timestamp of 'Apr 5 18:00'. The terminal shows the command 'tar -xzf data.tar.gz' being executed, followed by 'ls'. The output of 'ls' lists the following files and directories: Backup, Desktop, Downloads, link1, New, Pictures, snap, test.zip, data.tar.gz, Documents, file1.txt, Music, newlab.sh, Public, Templates, and Videos. The prompt returns to '\$' after the commands.

**Explanation**

The tar -xzf command is used to extract a compressed archive file.

Here, the file data.tar.gz is extracted, and the contents (such as the Music directory) are restored in the current directory.

The -x option extracts the archive, -z handles gzip compression, and -f specifies the file name.

The ls command confirms that the extraction was successful

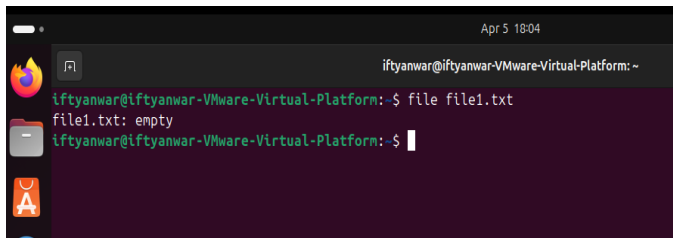
#### 44 Command: file

**Use Case:** Show file type

**Syntax:**

`file file1.txt`

**Output :**

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a timestamp of 'Apr 5 18:04'. The terminal shows the command 'file file1.txt' being executed. The output is 'file1.txt: empty'. The prompt returns to '\$'.

**Explanation**

The file command is used to determine the type of a file.

Here, it checks file1.txt and shows that the file is empty.

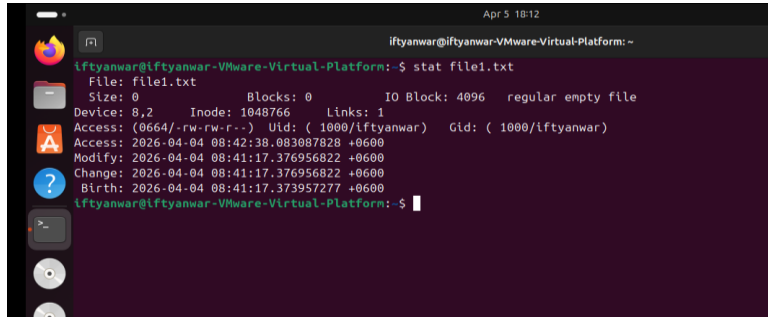
This command does not rely on file extension; it analyzes the actual content of the file.

#### 45 Command: stat

##### Syntax:

stat file1.txt

##### Output:



```
Apr 5 18:12
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ stat file1.txt
File: file1.txt
Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,2      Inode: 1048766   Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/iftyanwar)   Gid: ( 1000/iftyanwar)
Access: 2026-04-04 08:42:38.083087828 +0600
Modify: 2026-04-04 08:41:17.376956822 +0600
Change: 2026-04-04 08:41:17.376956822 +0600
Birth: 2026-04-04 08:41:17.373957277 +0600
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

##### Explanation:

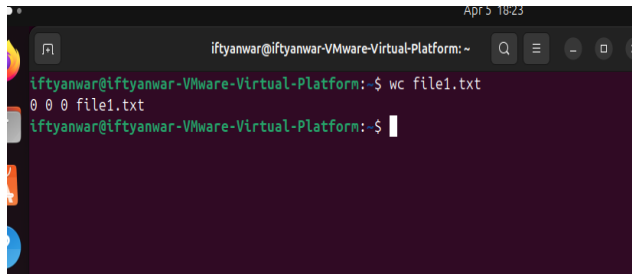
The stat command is used to display detailed information about a file. It shows file size, permissions, owner, group, and timestamps such as access, modify, and change time. In this case, file1.txt is an empty file with its metadata displayed

#### 46 Command: wc

##### Syntax:

wc file1.txt

##### Output:



```
Apr 5 18:23
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ wc file1.txt
0 0 0 file1.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

##### Explanation:

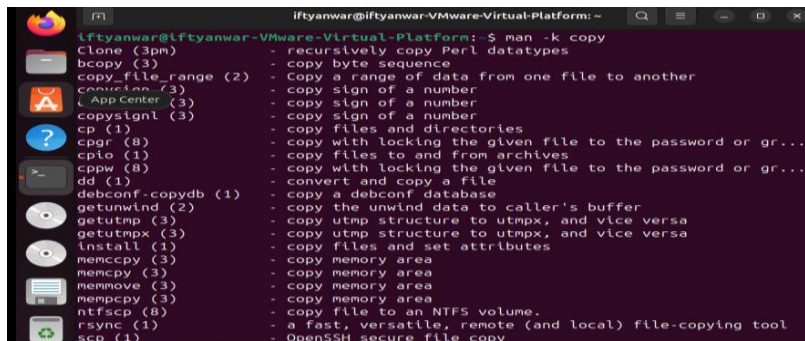
The wc command is used to count the number of lines, words, and characters in a file. In this case, file1.txt is empty, so it shows 0 lines, 0 words, and 0 characters.

#### 47 Command: man -k

##### Syntax:

man -k copy

##### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ man -k copy
Clone (3pm)      - recursively copy Perl datatypes
bcopy (3)        - copy byte sequence
copy_file_range (2) - Copy a range of data from one file to another
cmmirion (73)    - copy sign of a number
AppCenter (3)    - copy sign of a number
copysignl (3)     - copy files and directories
cp (1)           - copy with locking the given file to the password or gr...
cpgr (8)         - copy files to and from archives
cpio (1)         - copy with locking the given file to the password or gr...
cppw (8)         - convert and copy a file
dd (1)           - copy a debconf database
debconf-copydb (1) - copy the unwind data to caller's buffer
getunwind (2)    - copy utmp structure to utmpx, and vice versa
getutmp (3)      - copy utmp structure to utmpx, and vice versa
install (1)      - copy files and set attributes
memccpy (3)      - copy memory area
memcpy (3)       - copy memory area
memmove (3)      - copy memory area
mempcpy (3)      - copy memory area
ntfscpy (8)      - copy file to an NTFS volume.
rsync (1)        - a fast, versatile, remote (and local) file-copying tool
scp (1)          - OpenSSH secure file copy
```

### Explanation:

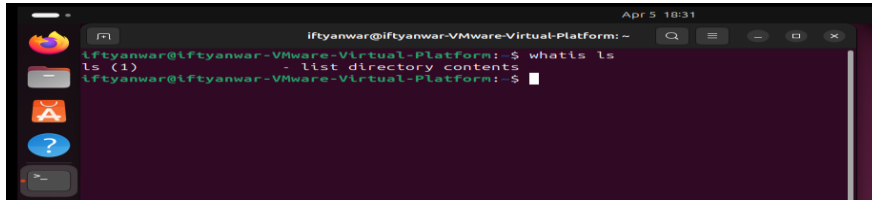
The man -k command is used to search the manual pages for a specific keyword. In this example, it searches for the keyword "copy" and displays a list of related commands along with their short descriptions.

**48 Command:** whatis

### Syntax:

whatis ls

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a date of 'Apr 5 18:31'. The prompt is '\$'. The command 'whatis ls' has been entered, and the output is 'ls (1) - list directory contents'. The prompt is now '\$' again.

### Explanation:

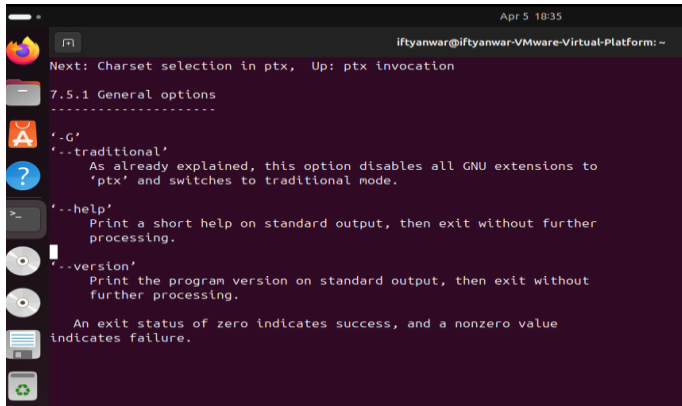
The whatis command is used to display a short description of a command. In this example, it shows that the ls command is used to list directory contents.

**49 Command:** info

### Syntax:

info ls

### Output

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a date of 'Apr 5 18:35'. The prompt is '\$'. The command 'info ls' has been entered, and the output shows the '7.5.1 General options' section of the ls manual page. It lists options like '-G', '--traditional', '--help', and '--version' with their descriptions. The prompt is now '\$' again.

### Explanation:

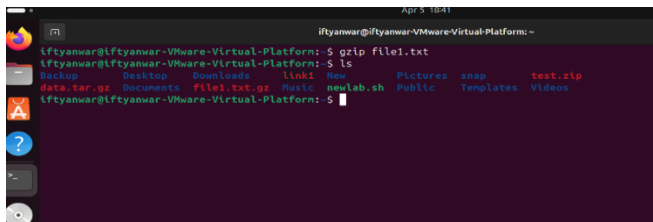
The info command is used to display detailed documentation of a command. It provides more comprehensive information compared to the man command and allows navigation through different sections.

**50 Command:** gzip

### Syntax:

gzip file1.txt

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a date of 'Apr 5 18:31'. The prompt is '\$'. The command 'gzip file1.txt' has been entered, and the output is 'gzip: file1.txt: OK'. The prompt is now '\$' again.

### Explanation:

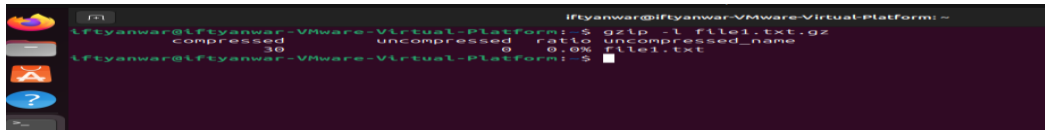
The gzip command is used to compress a file. It replaces the original file with a compressed version having the .gz extension. In this example, file1.txt is compressed into file1.txt.gz.

**51 Command:** gzip -l

### Syntax:

gzip -l file1.txt.gz

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ gzip -l file1.txt.gz  
compressed      uncompressed    ratio uncompressed_name  
0               0               0.0% file1.txt
```

### Explanation:

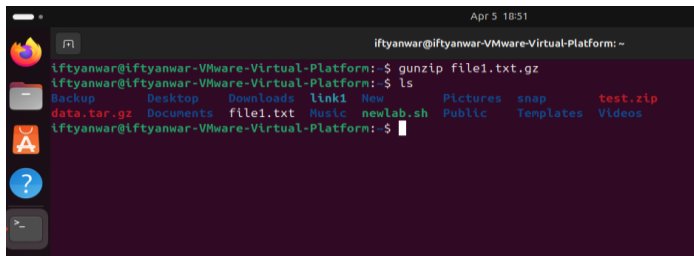
The gzip -l command is used to display information about a compressed file. It shows the compressed size, original size, compression ratio, and original file name.

**52 Command:** gunzip

### Syntax:

gunzip file1.txt.gz

### Output:



```
Apr 5 18:51  
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ gunzip file1.txt.gz  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls  
Backup Desktop Downloads link1 New Pictures snap test.zip  
data.tar.gz Documents file1.txt Music newLab.sh Public Templates Videos  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

### Explanation:

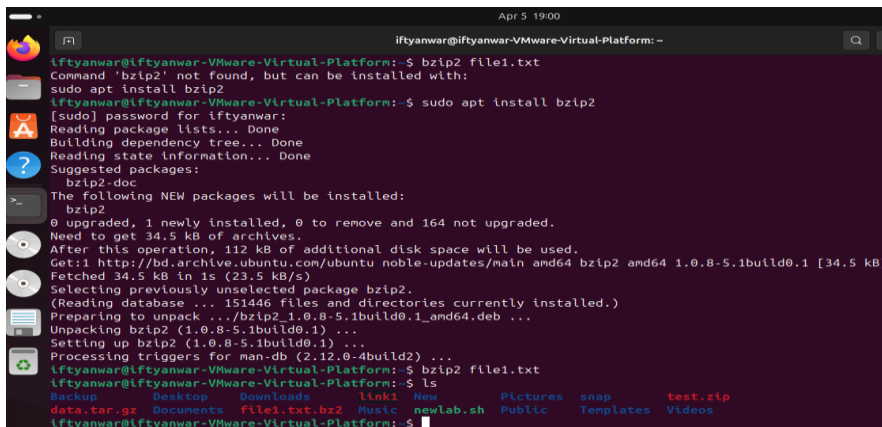
The gunzip command is used to decompress a .gz file. It restores the original file and removes the compressed file. In this example, file1.txt.gz is decompressed and the original file1.txt is restored successfully.

**53 Command:** bzip2

### Syntax:

bzip2 file1.txt

### Output:



```
Apr 5 19:00  
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ bzip2 file1.txt  
Command 'bzip2' not found, but can be installed with:  
sudo apt install bzip2  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ sudo apt install bzip2  
[sudo] password for iftyanwar:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Suggested packages:  
  bzip2-doc  
The following NEW packages will be installed:  
  bzip2  
0 upgraded, 1 newly installed, 0 to remove and 164 not upgraded.  
Need to get 34.5 kB of archives.  
After this operation, 112 kB of additional disk space will be used.  
Get:1 http://bd.archive.ubuntu.com/ubuntu noble-updates/main amd64 bzip2 amd64 1.0.8-5.1build0.1 [34.5 kB]  
Fetched 34.5 kB in 1s (23.5 kB/s)  
Selecting previously unselected package bzip2.  
(Reading database ... 151446 files and directories currently installed.)  
Preparing to unpack .../bzip2_1.0.8-5.1build0.1_amd64.deb ...  
Unpacking bzip2 (1.0.8-5.1build0.1) ...  
Setting up bzip2 (1.0.8-5.1build0.1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ bzip2 file1.txt  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls  
Backup Desktop Downloads link1 New Pictures snap test.zip  
data.tar.gz Documents file1.txt.bz2 Music newLab.sh Public Templates Videos  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

### Explanation:

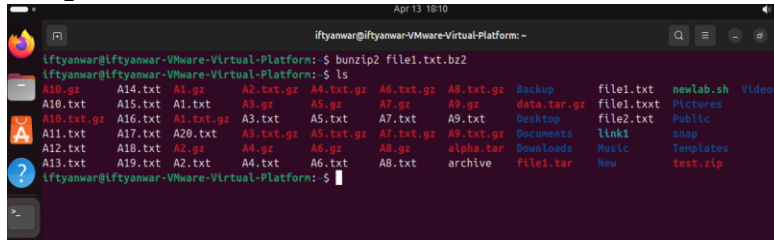
The bzip2 command is used to compress a file. It replaces the original file with a compressed version having the .bz2 extension. In this example, file1.txt is compressed into file1.txt.bz2.

**54 Command:** bzip2

### Syntax:

bunzip2 file1.txt.bz2

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' shows the command 'bunzip2 file1.txt.bz2' being executed. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~\$'. Below the command, the output of the 'ls' command is displayed, showing a list of files and directories including A10.gz, A14.txt, A1.gz, A2.txt.gz, A4.txt.gz, A6.txt.gz, A8.txt.gz, Backup, file1.txt, newlab.sh, Videos, A10.txt, A15.txt, A1.txt, A3.gz, A5.gz, A7.gz, A9.gz, data.tar.gz, file1.txt, Pictures, A10.txt.gz, A16.txt, A1.txt.gz, A3.txt, A5.txt, A7.txt, A9.txt, Desktop, file2.txt, Public, A11.txt, A17.txt, A20.txt, A3.txt.gz, A5.txt.gz, A7.txt.gz, A9.txt.gz, Documents, link1, snap, A12.txt, A18.txt, A2.gz, A4.gz, A6.gz, A8.gz, alpha.tar, Downloads, Music, Templates, A13.txt, A19.txt, A2.txt, A4.txt, A6.txt, A8.txt, archive, file1.tar, New, and test.zip.

### Explanation:

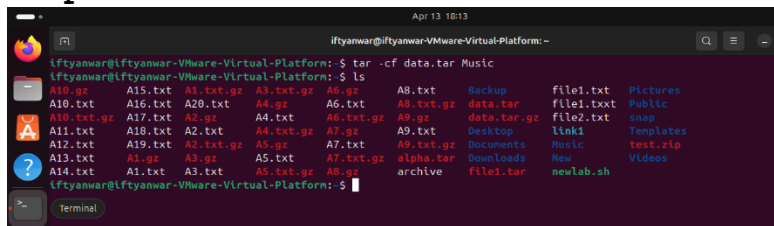
The bunzip2 command is used to decompress a .bz2 file. It restores the original file and removes the compressed file. In this example, file1.txt.bz2 is decompressed back to file1.txt.

**55 Command:** tar -cf

### Syntax:

tar -cf data.tar Music

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' shows the command 'tar -cf data.tar Music' being executed. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~\$'. Below the command, the output of the 'ls' command is displayed, showing a list of files and directories including A10.gz, A15.txt, A1.txt.gz, A3.txt.gz, A6.gz, A8.txt, Backup, file1.txt, Pictures, A10.txt, A16.txt, A20.txt, A4.gz, A6.txt, A8.txt.gz, data.tar, file1.txt, Public, A10.txt.gz, A17.txt, A2.gz, A4.txt, A6.txt.gz, A9.gz, data.tar.gz, file2.txt, snap, A11.txt, A18.txt, A2.txt, A4.txt.gz, A7.gz, A9.txt, Desktop, link1, Templates, A12.txt, A19.txt, A2.txt.gz, A5.gz, A7.txt, A9.txt.gz, Documents, Music, test.zip, A13.txt, A14.txt, A1.gz, A3.gz, A5.txt, A7.txt.gz, alpha.tar, Downloads, New, Videos, and file1.tar.

### Explanation:

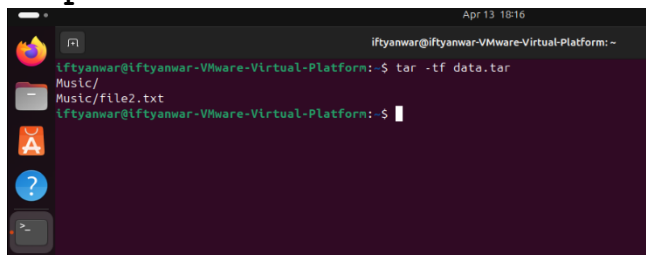
The tar -cf command is used to create an archive file. It combines files or directories into a single .tar file without compression. In this example, the Music directory is archived into data.tar.

**56 Command:** tar -tf

### Syntax:

tar -tf data.tar

### Output:

A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' shows the command 'tar -tf data.tar' being executed. The prompt is 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~\$'. Below the command, the output of the 'tar -tf data.tar' command is displayed, showing the contents of the data.tar archive, which include 'Music/' and 'Music/file2.txt'.

### Explanation:

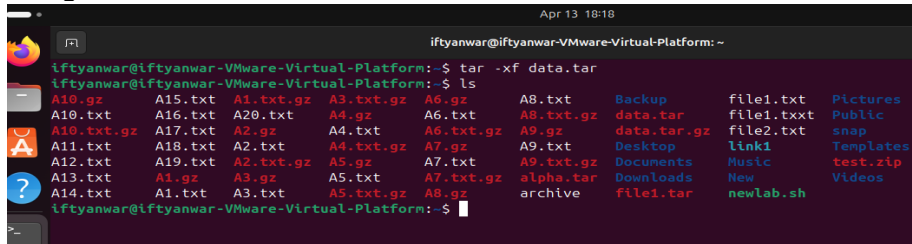
The tar -tf command is used to display the contents of a .tar archive without extracting it. It shows all files and directories stored inside the archive. In this example, it lists the contents of data.tar.

**57 Command:** tar -xf

**Syntax:**

tar -xf data.tar

**Output:**



```
Apr 13 18:18
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ tar -xf data.tar
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls
A10.gz  A15.txt  A1.txt.gz  A3.txt.gz  A6.gz  A8.txt  Backup  file1.txt  Pictures
A10.txt A16.txt  A20.txt   A4.gz     A6.txt  A8.txt.gz  data.tar  file1.txt  Public
A10.txt.gz A17.txt  A2.gz    A4.txt    A6.txt.gz  A9.gz  data.tar.gz  file2.txt  snap
A11.txt  A18.txt  A2.txt   A4.txt.gz  A7.gz    A9.txt  Desktop  link1     Templates
A12.txt  A19.txt  A2.txt.gz A5.gz     A7.txt    A9.txt.gz  Documents  Music    test.zip
A13.txt  A1.gz   A3.gz    A5.txt    A7.txt.gz alpha.tar  Downloads  New      Videos
A14.txt  A1.txt  A3.txt   A5.txt.gz A8.gz    archive  file1.tar  newlab.sh
```

**Explanation:**

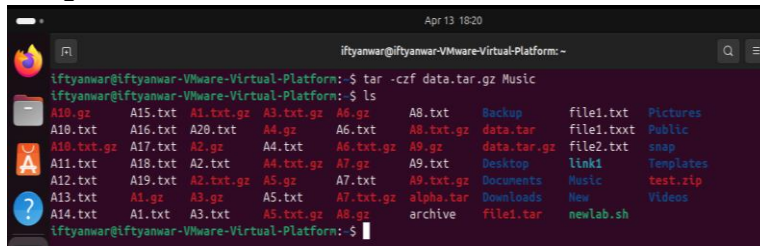
The tar -xf command is used to extract files from a .tar archive. It restores all files and directories from the archive into the current directory. In this example, the contents of data.tar are extracted.

**58 Command:** tar -czf

**Syntax:**

tar -czf data.tar.gz Music

**Output:**



```
Apr 13 18:20
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ tar -czf data.tar.gz Music
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls
A10.gz  A15.txt  A1.txt.gz  A3.txt.gz  A6.gz  A8.txt  Backup  file1.txt  Pictures
A10.txt A16.txt  A20.txt   A4.gz     A6.txt  A8.txt.gz  data.tar  file1.txt  Public
A10.txt.gz A17.txt  A2.gz    A4.txt    A6.txt.gz  A9.gz  data.tar.gz  file2.txt  snap
A11.txt  A18.txt  A2.txt   A4.txt.gz  A7.gz    A9.txt  Desktop  link1     Templates
A12.txt  A19.txt  A2.txt.gz A5.gz     A7.txt    A9.txt.gz  Documents  Music    test.zip
A13.txt  A1.gz   A3.gz    A5.txt    A7.txt.gz alpha.tar  Downloads  New      Videos
A14.txt  A1.txt  A3.txt   A5.txt.gz A8.gz    archive  file1.tar  newlab.sh
```

**Explanation:**

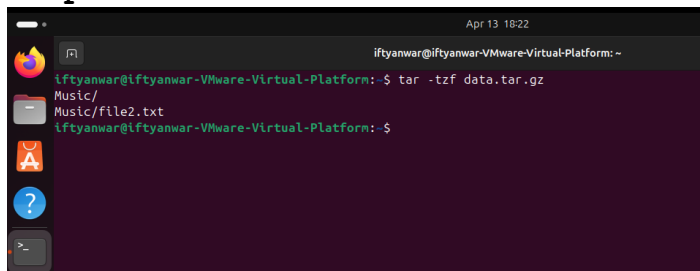
The tar -czf command is used to create a compressed archive using gzip. It combines and compresses files or directories into a single .tar.gz file. In this example, the Music directory is archived and compressed into data.tar.gz.

**59 Command:** tar -tzf

**Syntax:**

tar -tzf data.tar.gz

**Output:**



```
Apr 13 18:22
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ tar -tzf data.tar.gz
Music/
Music/file2.txt
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

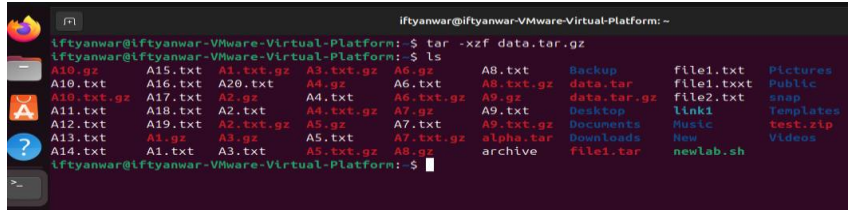
The tar -tzf command is used to display the contents of a compressed .tar.gz archive without extracting it. It shows all files and directories stored inside the archive. In this example, it lists the contents of data.tar.gz.

**60 Command:** tar -xzf

**Syntax:**

tar -xzf data.tar.gz

**Output:**



**Explanation:**

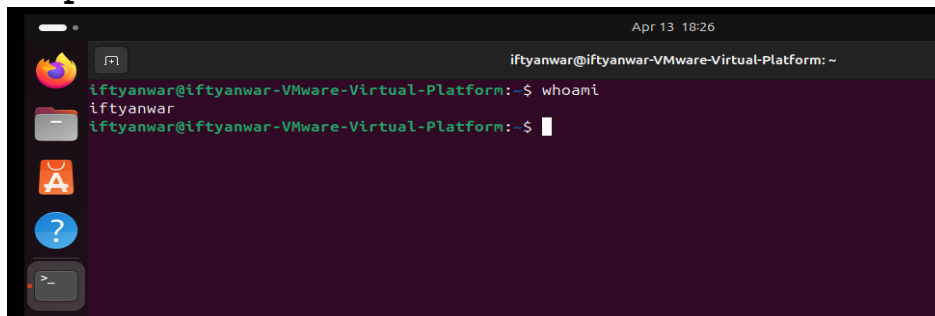
The tar -xzf command is used to extract a compressed .tar.gz archive. It restores all files and directories from the archive into the current directory. In this example, the contents of data.tar.gz are extracted successfully.

**61 Command:** whoami

**Syntax:**

whoami

**Output:**



**Explanation:**

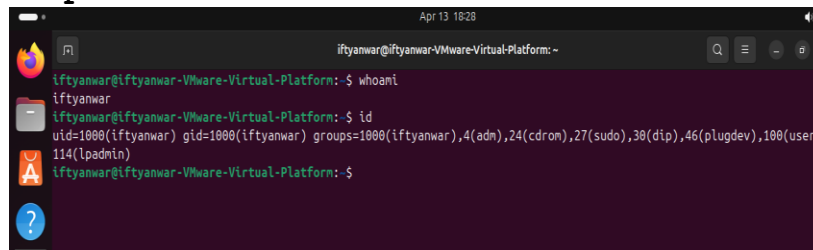
The whoami command is used to display the current logged-in user. It shows the username of the person who is currently using the system.

**62 Command:** id

**Syntax:**

id

**Output:**



**Explanation:**

The id command is used to display the user identity information. It shows the user ID (UID), group ID (GID), and the groups the user belongs to.

### 63 Command: who

#### Syntax:

who



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar$ whoami  
iftyanwar  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ id  
uid=1000(iftyanwar) gid=1000(iftyanwar) groups=1000(iftyanwar),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),  
114(lpadmin)  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ who  
iftyanwar  pts/0    2026-04-13 18:08 (login screen)  
iftyanwar  tty2     2026-04-13 18:08 (tty2)  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $
```

#### Explanation:

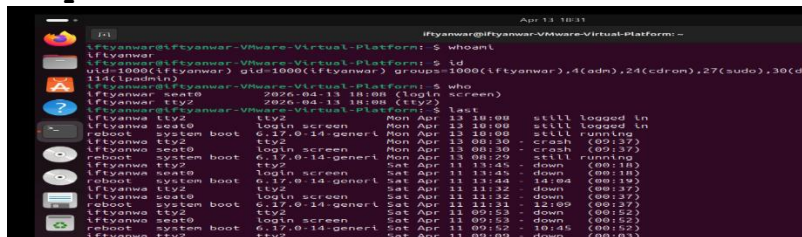
The who command is used to display information about users who are currently logged into the system. It shows the username, terminal, and login time.

### 64 Command: last

#### Syntax:

last

#### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar$ whoami  
iftyanwar  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ id  
uid=1000(iftyanwar) gid=1000(iftyanwar) groups=1000(iftyanwar),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),  
114(lpadmin)  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ who  
iftyanwar  pts/0    2026-04-13 18:08 (login screen)  
iftyanwar  tty2     2026-04-13 18:08 (tty2)  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $ last  
iftyanwar  tty2          tty2          Mon Apr 13 18:08      still logged in  
iftyanwar  pts/0          pts/0          Mon Apr 13 18:08      still logged in  
reboot     system boot    6.17.0-14-genert Mon Apr 13 18:08      still running  
iftyanwar  tty2          tty2          Mon Apr 13 08:30      crash (00:37)  
reboot     system boot    6.17.0-14-genert Mon Apr 13 08:29      still running  
iftyanwar  tty2          tty2          Sat Apr 11 13:45      down (00:18)  
iftyanwar  pts/0          pts/0          Sat Apr 11 13:45      down (00:18)  
reboot     system boot    6.17.0-14-genert Sat Apr 11 13:44      14.84 (00:19)  
iftyanwar  tty2          tty2          Sat Apr 11 11:32      down (00:37)  
iftyanwar  pts/0          pts/0          Sat Apr 11 11:32      down (00:37)  
reboot     system boot    6.17.0-14-genert Sat Apr 11 11:31      12:09 (00:27)  
iftyanwar  tty2          tty2          Sat Apr 11 09:53      down (00:52)  
iftyanwar  pts/0          pts/0          Sat Apr 11 09:53      down (00:52)  
reboot     system boot    6.17.0-14-genert Sat Apr 11 09:52      10:45 (00:52)  
iftyanwar  tty2          tty2          Sat Apr 11 09:09      down (00:43)
```

#### Explanation:

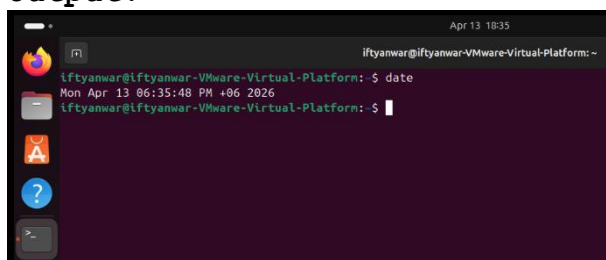
The last command is used to display the login history of users. It shows previous login sessions, system reboots, and logout times. This helps in tracking user activity on the system.

### 65 Command: date

#### Syntax:

date

#### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar$ date  
Mon Apr 13 06:35:48 PM +06 2026  
iftyanwar@iftyanwar-VMware-Virtual-Platform: $
```

#### Explanation:

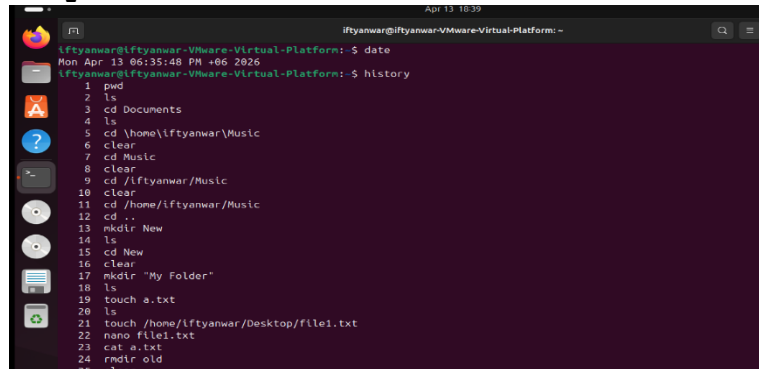
The date command is used to display the current system date and time. It shows the day, month, time, timezone, and year

## 66 Command: history

### Syntax:

history

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$ date
Mon Apr 13 06:35:48 PM +06 2026
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$ history
1  pwd
2  ls
3  cd Documents
4  ls
5  cd /home/iftyanwar/Music
6  clear
7  cd Music
8  clear
9  cd /iftyanwar/Music
10 clear
11 cd /home/iftyanwar/Music
12 cd ..
13 mkdir New
14 ls
15 cd New
16 clear
17 mkdir "My Folder"
18 ls
19 touch a.txt
20 ls
21 touch /home/iftyanwar/Desktop/file1.txt
22 nano file1.txt
23 cat a.txt
24 rm dir old
25 clear
```

### Explanation:

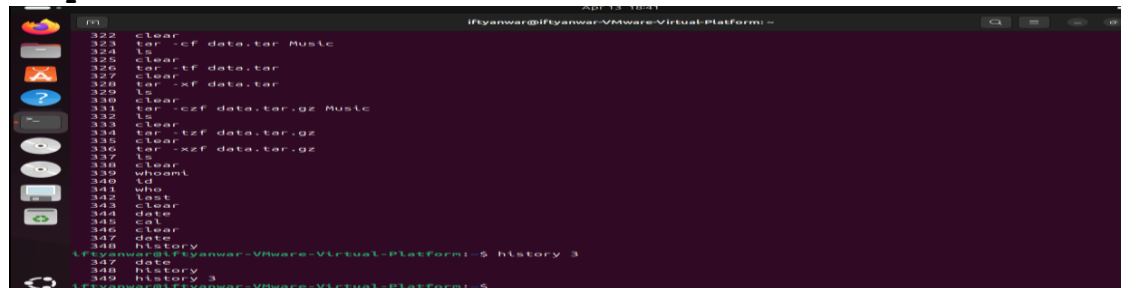
The history command shows previously executed commands.

## 67 Command: history3

### Syntax:

history3

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~$ history 3
322 clear
323 tar -czf data.tar.gz Music
324 ls
```

### Explanation:

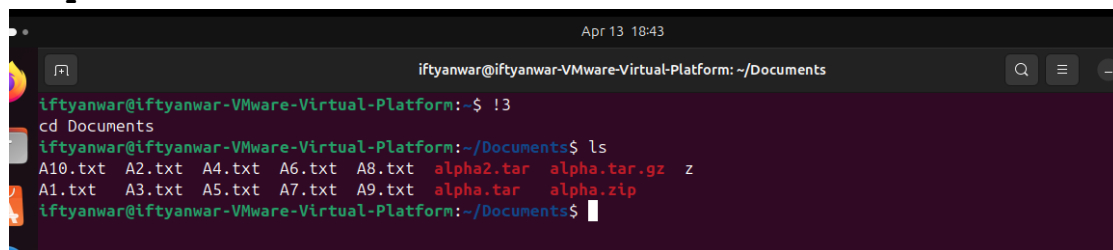
Displays the last 3 commands from the command history.

## 68 Command: !3

### Syntax:

!3

### Output:



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/Documents$ !3
cd Documents
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~/Documents$ ls
A10.txt A2.txt A4.txt A6.txt A8.txt alpha2.tar alpha.tar.gz z
A1.txt A3.txt A5.txt A7.txt A9.txt alpha.tar alpha.zip
```

### Explanation:

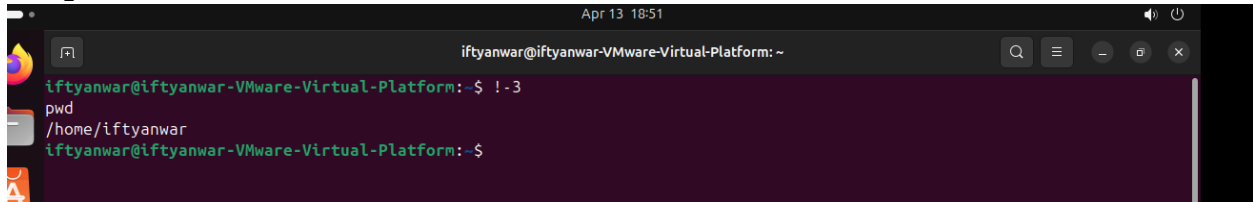
Runs a specific command from history using its number.

**69 Command:** `!-3`

**Syntax:**

`!-3`

**Output:**



```
Apr 13 18:51
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ !-3
pwd
/home/iftyanwar
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

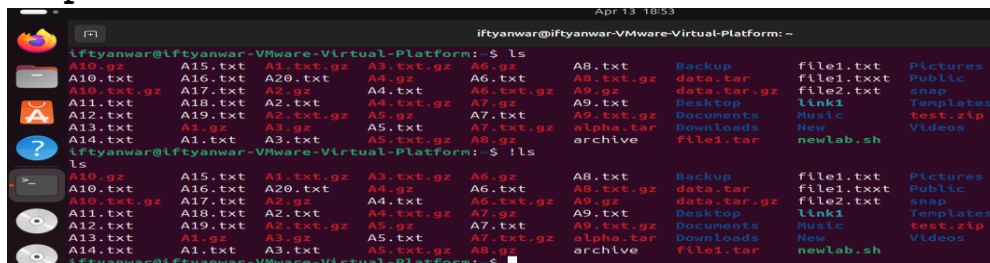
The `!-3` command is used to execute the command that was run three steps before. It must be written without any space between `!` and `-3`. Otherwise, it will produce an error.

**70 Command:** `!!s`

**Syntax:**

`!!s`

**Output:**



```
Apr 13 18:53
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls
A10.gz  A15.txt  A1.txt.gz  A3.txt.gz  A6.gz  A8.txt  Backup  file1.txt  Pictures
A10.txt  A16.txt  A20.txt  A4.gz  A6.txt  A8.txt.gz  data.tar  file1.txtxt  Public
A10.txt.gz  A17.txt  A2.gz  A4.txt  A9.gz  A9.txt  data.tar.gz  file2.txt  snap
A11.txt  A18.txt  A2.txt  A4.txt.gz  A7.gz  A9.txt  Desktop  link1  Templates
A12.txt  A19.txt  A2.txt.gz  A5.gz  A7.txt  A9.txt.gz  Documents  Music  test.zip
A13.txt  A1.gz  A3.gz  A5.txt  A7.txt.gz  alpha.tar  Downloads  New  Videos
A14.txt  A1.txt  A3.txt  A5.txt.gz  A8.gz  archive  file1.tar  newlab.sh
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ !!s
ls
A10.gz  A15.txt  A1.txt.gz  A3.txt.gz  A6.gz  A8.txt  Backup  file1.txt  Pictures
A10.txt  A16.txt  A20.txt  A4.gz  A6.txt  A8.txt.gz  data.tar  file1.txtxt  Public
A10.txt.gz  A17.txt  A2.gz  A4.txt  A9.gz  A9.txt  data.tar.gz  file2.txt  snap
A11.txt  A18.txt  A2.txt  A4.txt.gz  A7.gz  A9.txt  Desktop  link1  Templates
A12.txt  A19.txt  A2.txt.gz  A5.gz  A7.txt  A9.txt.gz  Documents  Music  test.zip
A13.txt  A1.gz  A3.gz  A5.txt  A7.txt.gz  alpha.tar  Downloads  New  Videos
A14.txt  A1.txt  A3.txt  A5.txt.gz  A8.gz  archive  file1.tar  newlab.sh
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

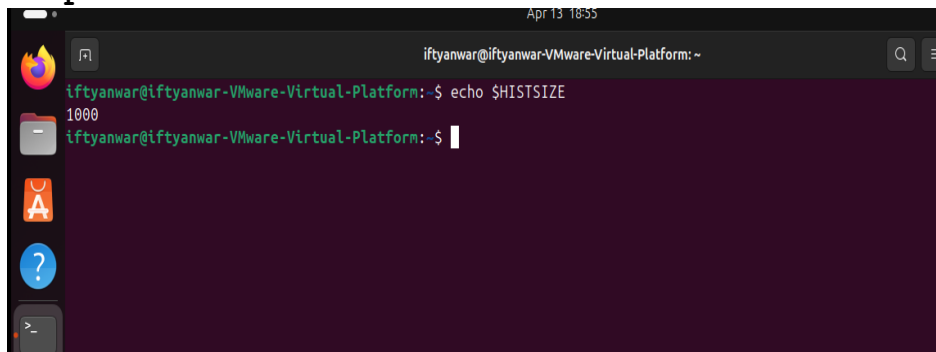
The `!!s` command is used to execute the most recent `ls` command from the command history. It reruns the last `ls` command and displays the same output.

**71 Command:** `echo $HISTSIZE`

**Syntax:**

`echo $HISTSIZE`

**Output:**



```
Apr 13 18:55
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ echo $HISTSIZE
1000
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

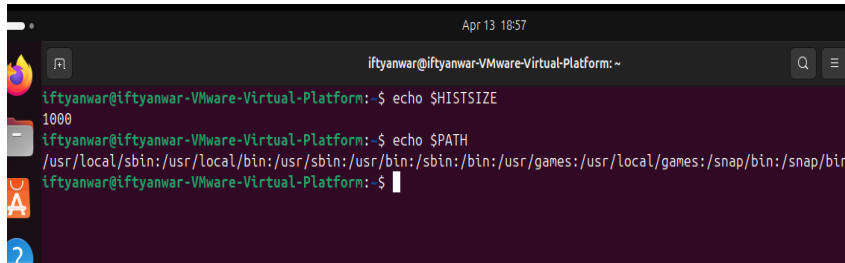
**Explanation:**

The echo \$HISTSIZE command is used to display the number of commands that can be stored in the history list. It shows the maximum size of the command history.

**72 Command:** echo \$PATH

**Syntax:**

echo \$PATH

**Output:**A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a search icon and a menu icon in the top right. The terminal shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ echo $HISTSIZE
1000
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

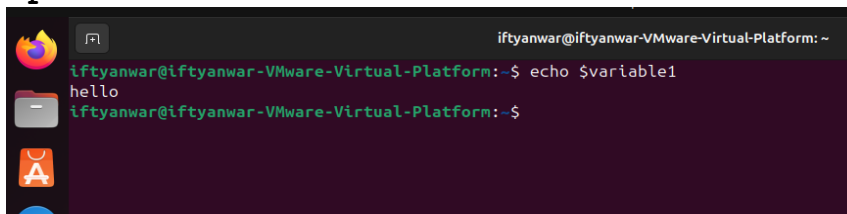
**Explanation:**

The echo \$PATH command is used to display the system's PATH variable. It shows the list of directories where the system looks for executable commands.

**73 Command:** export variable1

**Syntax:**

export variable1=hello

**Output:**A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a search icon and a menu icon in the top right. The terminal shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ echo $variable1
hello
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

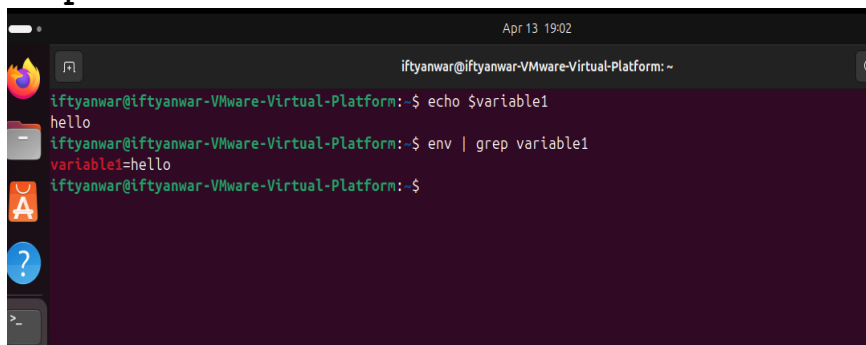
**Explanation:**

The export command is used to create an environment variable. In this example, a variable named variable1 is assigned the value "hello" and made available to the system.

**74 Command:** env | grep variable1

**Syntax:**

env | grep variable1

**Output:**A terminal window titled 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~' with a search icon and a menu icon in the top right. The terminal shows the following commands and output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ echo $variable1
hello
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ env | grep variable1
variable1=hello
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

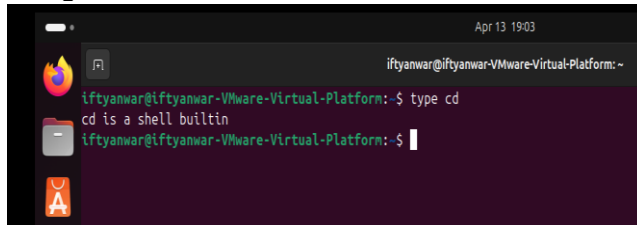
**Explanation:**

The `env | grep variable1` command is used to search for a specific environment variable. It displays the value of `variable1` from the list of environment variables.

**75 Command:** `type cd`

**Syntax:**

`type cd`

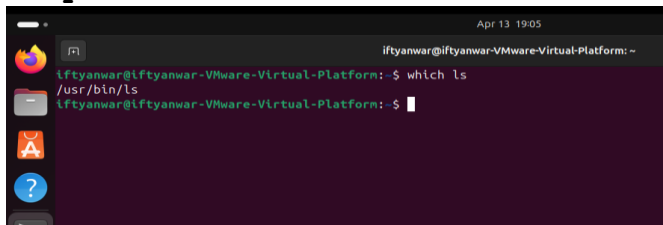
**Output:**A terminal window titled 'Apr 13 19:03' with the prompt 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~'. The user enters the command 'type cd'. The output shows 'cd is a shell builtin' followed by a new prompt line 'iftyanwar@iftyanwar-VMware-Virtual-Platform: \$'.**Explanation:**

The `type` command is used to display how a command is interpreted by the shell. In this case, it shows that `cd` is a shell builtin command, meaning it is built into the shell and not an external program.

**76 Command:** `which ls`

**Syntax:**

`which ls`

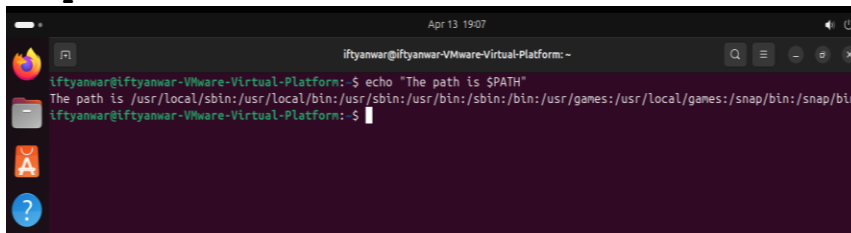
**Output:**A terminal window titled 'Apr 13 19:05' with the prompt 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~'. The user enters the command 'which ls'. The output shows '/usr/bin/ls' followed by a new prompt line 'iftyanwar@iftyanwar-VMware-Virtual-Platform: \$'.**Explanation:**

The `which` command is used to locate the path of an executable command. In this example, it shows that the `ls` command is located in `/usr/bin/ls`.

**77 Command:** `echo "The path is $PATH"`

**Syntax:**

`echo "The path is $PATH"`

**Output:**A terminal window titled 'Apr 13 19:07' with the prompt 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~'. The user enters the command 'echo "The path is \$PATH"'. The output shows 'The path is /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin' followed by a new prompt line 'iftyanwar@iftyanwar-VMware-Virtual-Platform: \$'.**Explanation:**

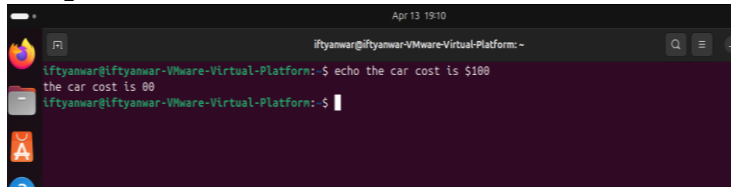
The `echo` command is used to display text on the terminal. In this example, it prints the value of the `PATH` variable along with a message. The `$PATH` variable contains a list of directories where executable commands are located.

**78 Command:** echo The car costs \$100

**Syntax:**

echo The car costs \$100

**Output:**



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ echo the car cost is $100  
the car cost is 00  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

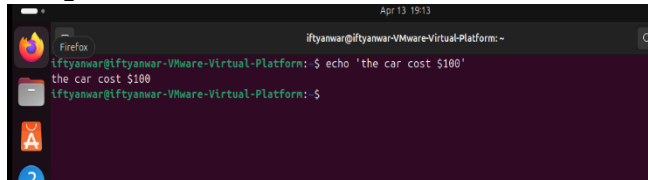
The echo command is used to display text. In this example, \$100 is treated as a variable, so \$1 is considered a variable and the remaining 00 is printed. Since the variable \$1 is empty, only 00 is shown in the output.

**79 Command:** echo 'The car costs \$100'

**Syntax:**

echo 'The car costs \$100'

**Output:**



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ echo 'the car cost $100'  
the car cost $100  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

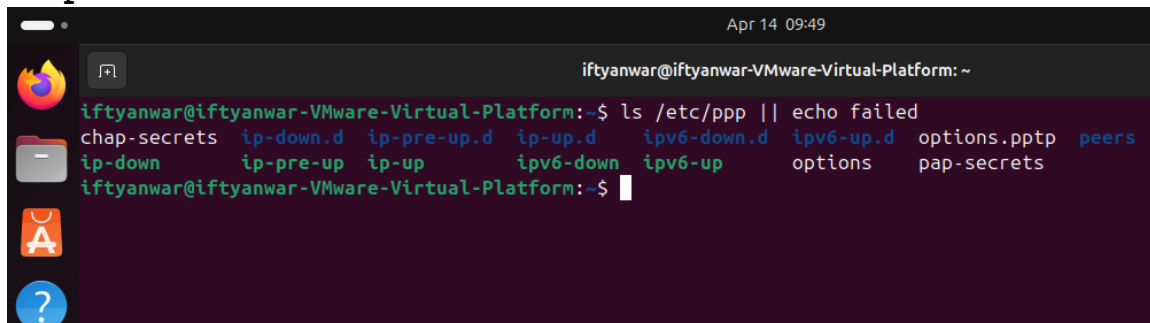
The echo command prints text on the terminal. When single quotes are used, special characters like \$ are not interpreted as variables. So the text is displayed exactly as written.

**80 Command:** ls/etc/ppp||echofailed

**Syntax:**

ls/etc/ppp||echofailed

**Output:**



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls /etc/ppp || echo failed  
chap-secrets  ip-down.d  ip-pre-up.d  ip-up.d     ipv6-down.d  ipv6-up.d    options.pptp  peers  
ip-down       ip-pre-up  ip-up        ipv6-down   ipv6-up      options      pap-secrets  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

The || operator runs the second command if the first command fails.

**81 Command:** ls /etc/ppp && echo success

**Syntax:**

ls /etc/ppp && echo success

**Output:**

```
Apr 14 09:52
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls /etc/ppp && echo success
chap-secrets  ip-down.d  ip-pre-up.d  ip-up.d      ipv6-down.d  ipv6-up.d    options.pptp  peers
ip-down       ip-pre-up  ip-up        ipv6-down    ipv6-up      options      pap-secrets
success
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

### Explanation:

The && operator runs the second command only if the first command is successful. In this example, if the /etc/ppp directory exists, it lists the contents and then prints "success".

**82 Command:** man ls

### Syntax:

man ls

### Output:

```
Apr 14 09:54
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
LS(1)
User Commands
NAME
ls - list directory contents
SYNOPSIS
ls [OPTION]... [FILE]...
DESCRIPTION
List information about the FILES (the current directory by default). Sort entries alphabetically if
-cftuvSUX nor --sort is specified.
Mandatory arguments to long options are mandatory for short options too.
-a, --all
do not ignore entries starting with .
-A, --almost-all
do not list implied . and ..
--author
with -l, print the author of each file
```

### Explanation:

The man command is used to display the manual page of a command. It provides detailed information about usage, options, and functionality of the ls command.

**83 Command:** whereis ls

### Syntax:

whereis ls

### Output:

```
Apr 14 09:57
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ man ls
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ whereis ls
ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

### Explanation:

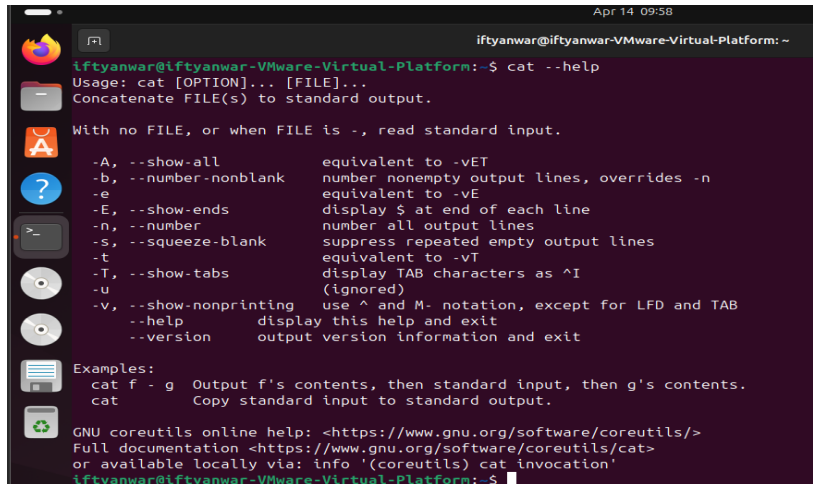
The whereis command is used to locate the binary, source, and manual files of a command. In this example, it shows the locations of the ls command and its manual page.

**84 Command:** cat --help

**Syntax:**

cat --help

**Output:**



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cat --help  
Usage: cat [OPTION]... [FILE]...  
Concatenate FILE(s) to standard output.  
  
With no FILE, or when FILE is -, read standard input.  
  
-A, --show-all           equivalent to -vET  
-b, --number-nonblank     number nonempty output lines, overrides -n  
-e, --show-ends           equivalent to -vE  
-E, --show-ends           display $ at end of each line  
-n, --number              number all output lines  
-s, --squeeze-blank       suppress repeated empty output lines  
-t, --show-tabs           equivalent to -vT  
-T, --show-tabs           display TAB characters as ^I  
-u, --show-nonprinting    use ^ and M- notation, except for LFD and TAB  
-v, --show-nonprinting    display this help and exit  
--help                   output version information and exit  
--version  
  
Examples:  
cat f - g Output f's contents, then standard input, then g's contents.  
cat      Copy standard input to standard output.  
  
GNU coreutils online help: <https://www.gnu.org/software/coreutils/>  
Full documentation <https://www.gnu.org/software/coreutils/cat>  
or available locally via: info '(coreutils) cat invocation'  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

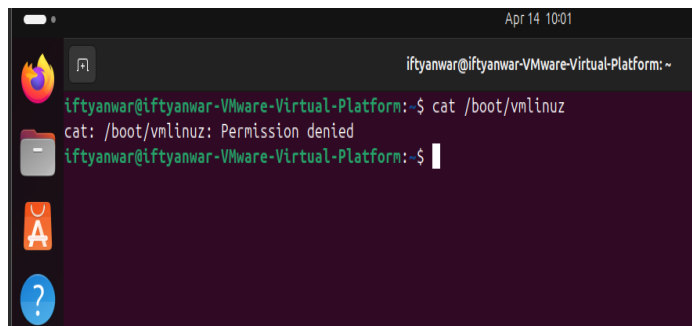
The cat --help command is used to display the usage information and available options of the cat command. It helps users understand how to use the command properly.

**85 Command:** cat /boot/vmlinuz

**Syntax:**

cat /boot/vmlinuz

**Output:**



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cat /boot/vmlinuz  
cat: /boot/vmlinuz: Permission denied  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

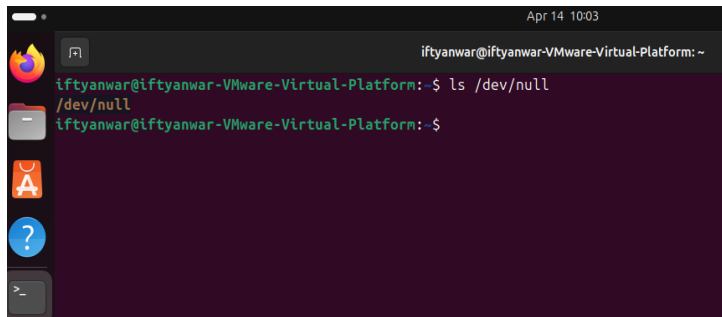
The /boot/vmlinuz file contains the Linux kernel. When viewed using cat, it displays unreadable binary data because it is not a text file but a compiled system file.

**86 Command:** ls /dev/null

**Syntax:**

ls /dev/null

**Output:**



```
Apr 14 10:03
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls /dev/null
/dev/null
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

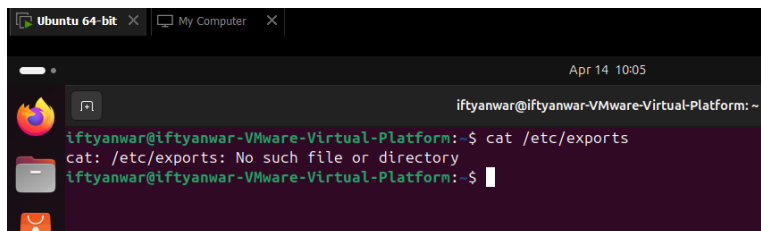
The /dev/null is a special device file that discards all data written to it. The ls /dev/null command shows that the file exists in the system.

**87 Command:** cat /etc/exports

**Syntax:**

cat /etc/exports

**Output:**



```
Apr 14 10:05
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cat /etc/exports
cat: /etc/exports: No such file or directory
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$
```

**Explanation:**

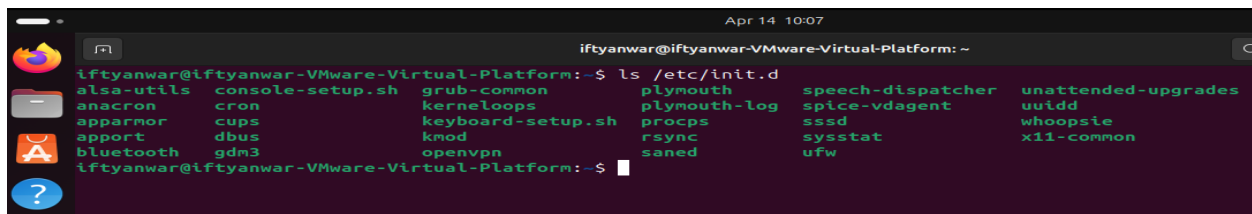
The /etc/exports file is used to configure shared directories for NFS (Network File System). The cat command displays its contents, which may be empty or contain configuration entries.

**88 Command:** ls /etc/init.d

**Syntax:**

ls /etc/init.d

**Output:**



```
Apr 14 10:07
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls /etc/init.d
alsa-utils  console-setup.sh  grub-common  plymouth  speech-dispatcher  unattended-upgrades
anacron     cron              kerneloops   plymouth-log  spice-vdagent      uidd
apparmor    cups             keyboard-setup.sh  procps      sssd               whoopsie
apport      dbus            kmod         rsync        sysstat            x11-common
bluetooth   gdm3            openvpn      saned        ufw
```

**Explanation:**

The /etc/init.d directory contains system service scripts. The ls command lists all available startup services in this directory.

**89 Command:** cat /etc/hosts

**Syntax:**

cat /etc/hosts

**Output:**

```
ifftanwar@ifftanwar-VMware-Virtual-Platform: ~  
ifftanwar@ifftanwar-VMware-Virtual-Platform:~$ cat /etc/hosts  
127.0.0.1 localhost  
127.0.1.1 ifftanwar-VMware-Virtual-Platform  
  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters  
ifftanwar@ifftanwar-VMware-Virtual-Platform:~$
```

### Explanation:

The /etc/hosts file contains mappings between IP addresses and hostnames. The cat command displays these mappings, which are used by the system to resolve hostnames locally.

**90 Command:** cat /etc/passwd

### Syntax:

cat /etc/passwd

### Output:

```
Apr 14 10:11  
ifftanwar@ifftanwar-VMware-Virtual-Platform: ~  
ifftanwar@ifftanwar-VMware-Virtual-Platform:~$ cat /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin  
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
```

### Explanation:

The /etc/passwd file contains information about all user accounts in the system. Each line represents a user with details such as username, user ID, group ID, home directory, and default shell

**91 Command:** cat /etc/profile

### Syntax:

cat /etc/profile

### Output:

```
ifftanwar@ifftanwar-VMware-Virtual-Platform: ~  
ifftanwar@ifftanwar-VMware-Virtual-Platform:~$ cat /etc/profile  
# /etc/profile: system-wide .profile file for the Bourne shell (sh(1))  
# and Bourne compatible shells (bash(1), ksh(1), ash(1), ...).  
  
if [ "$PS1" ]; then  
  if [ "$BASH" ] && [ "$BASH" != "/bin/sh" ]; then  
    # The file bash.bashrc already sets the default PS1.  
    # PS1="\h:\w\$ "  
    if [ -f /etc/bash.bashrc ]; then  
      . /etc/bash.bashrc  
    fi  
  else  
    if [ "${id -u}" -eq 0 ]; then  
      PS1="# "  
    else  
      PS1="$ "  
    fi  
  fi  
fi  
  
if [ -d /etc/profile.d ]; then  
  for i in /etc/profile.d/*.sh; do  
    if [ -r $i ]; then  
      . $i  
    fi  
  done  
unset i  
fi  
ifftanwar@ifftanwar-VMware-Virtual-Platform:~$
```

### Explanation:

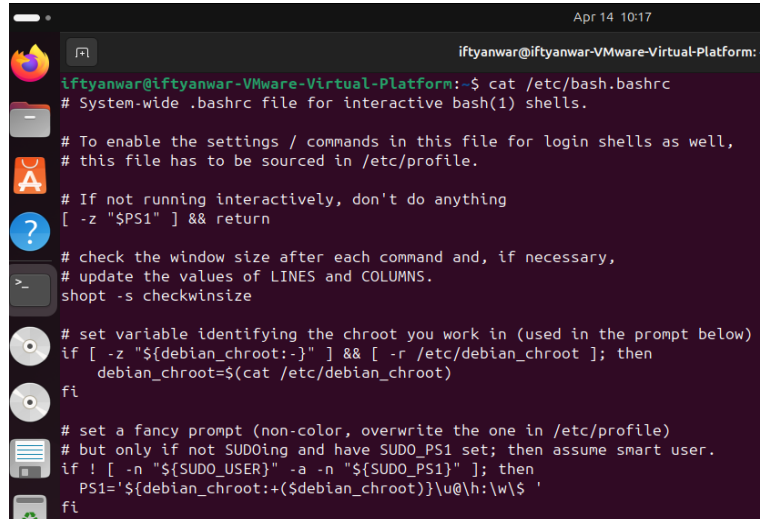
The /etc/profile file contains system-wide environment and startup settings. It is executed when a user logs into the system, setting variables and configurations for all users.

**92 Command:** cat /etc/profile

**Syntax:**

cat /etc/profile

**Output:**



```
Apr 14 10:17
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cat /etc/bash.bashrc
# System-wide .bashrc file for interactive bash(1) shells.

# To enable the settings / commands in this file for login shells as well,
# this file has to be sourced in /etc/profile.

# If not running interactively, don't do anything
[ -z "$PS1" ] && return

# check the window size after each command and, if necessary,
# update the values of LINES and COLUMNS.
shopt -s checkwinsize

# set variable identifying the chroot you work in (used in the prompt below)
if [ -z "${debian_chroot:-}" ] && [ -r /etc/debian_chroot ]; then
    debian_chroot=$(cat /etc/debian_chroot)
fi

# set a fancy prompt (non-color, overwrite the one in /etc/profile)
# but only if not SUDOing and have SUDO_PS1 set; then assume smart user.
if ! [ -n "${SUDO_USER}" -a -n "${SUDO_PS1}" ]; then
    PS1='${debian_chroot:+($debian_chroot)}\u@\h:\w\$ '
fi
```

**Explanation:**

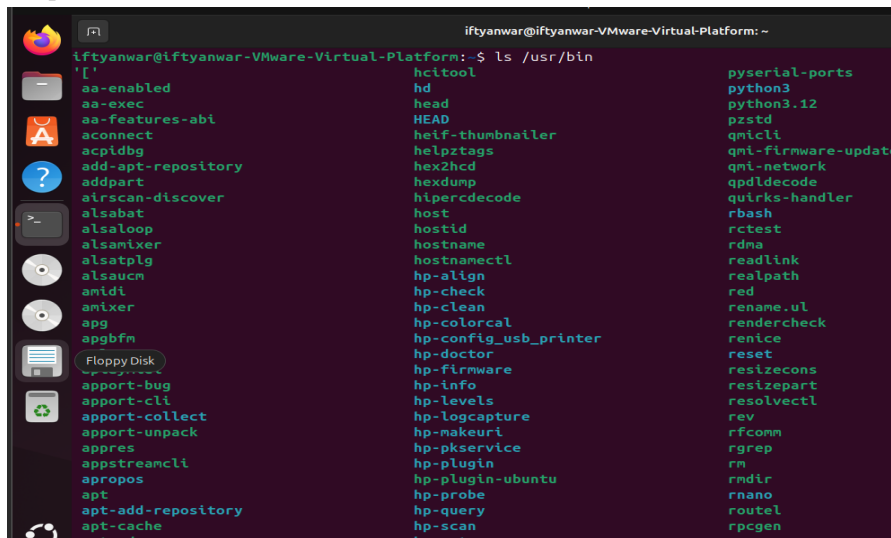
The /etc/profile file contains system-wide environment settings. However, in some Ubuntu systems, this file may not exist or may be replaced by other configuration files such as /etc/bash.bashrc or /etc/environment.

**93 Command:** ls /usr/bin

**Syntax:**

ls /usr/bin

**Output**



```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ ls /usr/bin
['
aa-enabled          hcitool             pyserial-ports
aa-exec             hd                   python3
aa-features-abi     head                 python3.12
aconnect            help2tags            pzsdt
acpidbg             hex2hcd              qnlcli
add-apt-repository  hexdump              qnl-firmware-update
addpart             hipercdecode          qnl-network
aircan-discover     host                 qpiddecode
alsabat             hostid                quirks-handler
alsaloop            hostname             rbash
alsamixer           hostnamectl           rctest
alsatplg            hp-align             rdma
alsaucm             hp-check              readlink
amidi               hp-clean              realpath
amixer              hp-colorcal           red
apg                 hp-config_usb_printer rename.ul
apgbfn              hp-doctor             rendercheck
apparmor            hp-firmware           renice
apparmor-bug        hp-info               reset
apparmor-cli        hp-levels             resizecons
apparmor-collect    hp-logcapture          resizepart
apparmor-unpack     hp-makeuri            resolvectl
appres              hp-pkcservice         rev
appstreamcli        hp-plugin              rfcomm
apropos              hp-plugin-ubuntu       rgrep
apt                  hp-probe              rm
apt-add-repository  hp-query              rmdir
apt-cache            hp-scan               rnano
apt-cdrom            hp-setup              routel
                    hp-setup              rpcgen
                    hp-setup              rsync
```

**Explanation:**

The /usr/bin directory contains most of the executable commands in the system. The ls command lists all available programs stored in this directory.



### Explanation:

The /usr/lib directory contains system libraries required by programs. These libraries help software run properly. The ls command lists all library files and folders in this directory.

**97 Command:** cat /proc/cpuinfo

### Syntax:

cat /proc/cpuinfo

### Output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cat /proc/cpuinfo  
processor       : 0  
vendor_id      : GenuineIntel  
cpu family     : 6  
model          : 154  
model name     : 12th Gen Intel(R) Core(TM) i3-1215U  
stepping       : 4  
microcode      : 0xffffffff  
cpu MHz        : 2496.001  
cache size     : 10240 KB  
physical id    : 0  
siblings       : 1  
core id        : 0  
cpu cores      : 1  
apicid         : 0  
initial apicid : 0  
fpu            : yes  
fpu_exception  : yes  
cpuid level    : 32  
wp             : yes  
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush  
yscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good noopl xtopology tsc_reliable nonstop  
req_pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand  
3dnowprefetch ptii ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed  
clwb sha_ni xsaveopt xsavec xgetbv1 xsavev avx_vnni arat unip gfni vaes vpclmulqdq rdpid movdiri mo  
serialize flush_lid arch_capabilities  
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs retbleed rfo  
its  
bogomips       : 4992.00  
clflush size   : 64  
cache_alignment : 64  
address sizes  : 45 bits physical, 48 bits virtual
```

### Explanation:

The /proc/cpuinfo file contains detailed information about the CPU. The cat command displays processor details such as model, speed, and cores.

**98 Command:** cat /proc/interrupts

### Syntax:

cat /proc/interrupts

### Output:

```
iftyanwar@iftyanwar-VMware-Virtual-Platform: ~  
iftyanwar@iftyanwar-VMware-Virtual-Platform:~$ cat /proc/interrupts  
CPU0      CPU1      CPU2      CPU3  
0:         27          0          0          0 IO-APIC 2-edge timer  
1:          0          0          0 920 IO-APIC 1-edge i8042  
6:          0          2          0          0 IO-APIC 6-edge floppy  
8:          0          0          0          0 IO-APIC 8-edge rtc0  
9:          0          0          0          0 IO-APIC 9-fasteoi acpi  
12:         0          0 16881          0 IO-APIC 12-edge i8042  
14:         0          0          0          0 IO-APIC 14-edge ata_piix  
15:         0          0          0          0 IO-APIC 15-edge ata_piix  
16:         0          0 5541          0 IO-APIC 16-fasteoi vmwgfx, snd_ens1371  
17:         0 9857          0          0 IO-APIC 17-fasteoi ehci_hcd:usb2, ioc0  
18:         67          0          0          0 IO-APIC 18-fasteoi uhci_hcd:usb1  
19:          0          0          0 3135 IO-APIC 19-fasteoi ens33  
24:         0          0          0          0 PCI-MSI-0000:00:15.0 0-edge PCIE PME, pci  
25:         0          0          0          0 PCI-MSI-0000:00:15.1 0-edge PCIE PME, pci  
26:         0          0          0          0 PCI-MSI-0000:00:15.2 0-edge PCIE PME, pci  
27:         0          0          0          0 PCI-MSI-0000:00:15.3 0-edge PCIE PME, pci  
28:         0          0          0          0 PCI-MSI-0000:00:15.4 0-edge PCIE PME, pci  
29:         0          0          0          0 PCI-MSI-0000:00:15.5 0-edge PCIE PME, pci  
30:         0          0          0          0 PCI-MSI-0000:00:15.6 0-edge PCIE PME, pci  
31:         0          0          0          0 PCI-MSI-0000:00:15.7 0-edge PCIE PME, pci  
32:         0          0          0          0 PCI-MSI-0000:00:16.0 0-edge PCIE PME, pci  
33:         0          0          0          0 PCI-MSI-0000:00:16.1 0-edge PCIE PME, pci  
34:         0          0          0          0 PCI-MSI-0000:00:16.2 0-edge PCIE PME, pci  
35:         0          0          0          0 PCI-MSI-0000:00:16.3 0-edge PCIE PME, pci  
36:         0          0          0          0 PCI-MSI-0000:00:16.4 0-edge PCIE PME, pci  
37:         0          0          0          0 PCI-MSI-0000:00:16.5 0-edge PCIE PME, pci  
38:         0          0          0          0 PCI-MSI-0000:00:16.6 0-edge PCIE PME, pci  
39:         0          0          0          0 PCI-MSI-0000:00:16.7 0-edge PCIE PME, pci  
40:         0          0          0          0 PCI-MSI-0000:00:17.0 0-edge PCIE PME, pci
```

### Explanation:

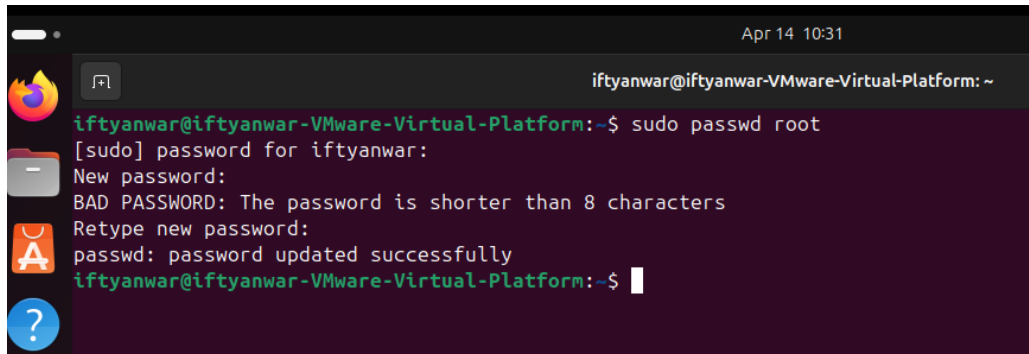
The /proc/interrupts file contains information about hardware interrupts in the system. The cat command displays how interrupts are handled by the CPU and devices

### 99 Command: sudo passwd root

#### Syntax:

sudo passwd root

#### Output:

A terminal window titled 'Apr 14 10:31' with the prompt 'iftyanwar@iftyanwar-VMware-Virtual-Platform: ~'. The user enters 'sudo passwd root'. The terminal shows the following output: '[sudo] password for iftyanwar:', 'New password:', 'BAD PASSWORD: The password is shorter than 8 characters', 'Retype new password:', 'passwd: password updated successfully', and the prompt returns to 'iftyanwar@iftyanwar-VMware-Virtual-Platform:~\$'.

#### Explanation:

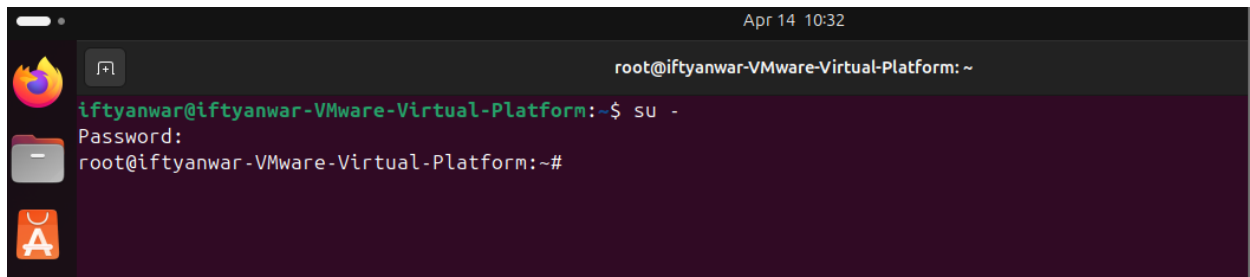
The sudo passwd root command is used to set or change the root user password. It allows enabling the root account by assigning a new password

### 100. Command: su -

#### Syntax:

su -

#### Output:

A terminal window titled 'Apr 14 10:32' with the prompt 'root@iftyanwar-VMware-Virtual-Platform: ~'. The user enters 'su -'. The terminal shows the following output: 'Password:', and the prompt changes to 'root@iftyanwar-VMware-Virtual-Platform:~#'.

#### Explanation:

The su - command is used to switch to the root user. After entering the correct root password, the user gains full administrative privileges and the prompt changes to root.

## Conclusion

This lab experiment provided a comprehensive understanding of Linux operating system commands and their practical applications. Throughout the lab, a wide range of commands were executed, including file and directory management, user management, system monitoring, compression techniques, and command-line utilities.

All commands were successfully practiced and verified in the Ubuntu environment. Both basic and advanced commands such as ls, cp, mv, chmod, grep, find, tar, gzip, and system-level commands were implemented to gain hands-on experience.

Additionally, system files and directories were explored to understand the internal structure of the Linux operating system. Command history, environment variables, and logical operators were also tested, enhancing overall command-line proficiency.

This lab significantly improved practical skills, problem-solving ability, and familiarity with Linux terminal operations. The knowledge gained from this experiment will be highly beneficial for future system administration and development tasks.

All required and additional commands were successfully executed and verified

## References

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<https://manpages.ubuntu.com>
2. GNU Coreutils Documentation  
<https://www.gnu.org/software/coreutils/manual>
3. Linux Tutorial – W3Schools  
<https://www.w3schools.com/linux>
4. Ubuntu Official Documentation  
<https://help.ubuntu.com>
5. Lab Manual provided by **Rashed Miah Sir**  
Lecturer, Department of Computer Science and Engineering  
Premier University, Chattogram